Table of Contents

1. INTRODUCTION 1
2. PROBLEM STATEMENT 2
3. OBJECTIVES 2
4. SCOPE OF THE PROJECT 2
5. LITERATURE REVIEW 3
6. METHODOLOGY 4
6.1 SYSTEM DEVELOPMENT MODEL 4
6.2SOFTWARE AND HARDWARE REQUIREMENT 5
6.3 USE CASE DIAGRAM 6
6.4 CONTEXT DIAGRAM 6
6.5 DATA FLOW DIAGRAM 7
6.6 TABLE RELATIONSHIP DIAGRAM 9
7. RESULTS AND DISCUSSION 10
8. CONCLUSION 38
9. FUTURE WORKS 38
10. GANTT CHART 39
11. REFERENCE 39

Abstract

This project introduces an Online Food Ordering System designed to make ordering meals from restaurants quicker, easier, and more convenient. Through a single digital platform, users can explore menus, place their orders, track their order status, and pay securely, without needing to visit the restaurant in person. On the other hand, restaurant staff can easily manage menus, process incoming orders, and keep track of daily operations in a more organized way.

The development process followed a structured software methodology, starting from gathering requirements and moving through system modeling, database design, and implementation. Key elements of the system include case diagrams, a context diagram, data flow diagrams, and a relational database setup that keeps information consistent and the system running smoothly. The solution is built using accessible hardware and software, making it practical and affordable to put into real-world use.

The results show that the system boosts order accuracy, cuts down waiting times, and offers smoother experience for customers, while also giving restaurants better control over their workflow. Overall, this project highlights how digital tools can simplify food service operations and lays the groundwork for future improvements like delivery tracking, analytics, and support for multiple restaurants.

Letter of Recommendation

This is to certify that Suresh Dahal, Sanskar Bhattarai, Dolindra Bahadur Raut**,** and Yugin Paudel**,** students of Bachelors in Information Technology (BIT), Patan Multiple Campus have successfully completed the project titled **"**Online Food Ordering System**"** under my supervision.

The group has shown strong teamwork, clear understanding of system analysis and design, and the ability to apply theoretical knowledge to a real working system. Their project involves designing and developing an online platform that allows users to browse menus, place food orders, and interact with restaurants digitally. They have also implemented administrative features that help restaurants manage menus, orders, and customer interactions efficiently.

Throughout the project, the team demonstrated solid skills in requirement analysis, system modeling, interface development, and problem-solving. Their coordination, division of work, and consistency in delivering results are commendable.

I am confident that each member of this group has the capability to excel in future academic and professional opportunities related to system analysis and design. I strongly recommend them for any role or project aligned with their skills.

Sincerely,

…….
Shiv Kumar Mahto
Lecturer of System Analysis and Design
Patan Multiple Campus
Tribhuvan University

1. INTRODUCTION

Online Food Ordering system is a process in which one can order various foods from some local restaurants and hotels. Nowadays everyone is having busy schedule whether it is urban areas or rural areas. People are so busy in their life that they don't get enough time to have their meals properly. If we talk about the children in the modern era, they like only fast food or something from the outside. They are ignoring eating homemade meals. So, food ordering system these days has one of the fastest growing markets, though being a new idea. In this project we will develop something like the same to help restaurants to get more exposure and increase revenue.

Our online food ordering system will have two components – a website that allows customers to view the menu and a place an order, and an admin as well as kitchen interface that enables the restaurant to receive and fulfill customer orders. The main purpose of an online ordering system is to provide customers with a way to place an order remotely by the internet. So why is this important? The main reason is that it benefits both the customer and the business. It allows kitchen staff to view order information. Some of the main reasons to build this system is that customers must put more effort into finding a specific food on traditional system. This system helps to overcome the effort for customers to find his/her favorite food. This system can be used as a tool for generating more profits and organizing food ordering processes in a better way. This system will allow restaurant owners to save on labor costs. This system will help a restaurant to get orders quickly and accurately.

2. PROBLEM STATEMENT

In the lack of an online food ordering system following issues may arise:

- Customers can get confused if they see an old print out of the menu instead of the new one
- It is hard for customers to find a specific food item on a paper-based menu
- It is hard for a restaurant manager to view sales records on the traditional system
- In a traditional paper-based menu, the food ingredients are not listed. So, customers had to seek other people's help to find it out

3. OBJECTIVES

The objectives of food ordering system are given below:

- To allow customers to order food online.
- To digitalize the menu of a restaurant.
- To allow admin to view sales records easily.

4. SCOPE OF THE PROJECT

- This system can be used in any restaurant or similar setting to digitalize their menu and simplify food ordering process.
- This system can allow customers to order variety of food items.

5. LITERATURE REVIEW

In the traditional system, customers had to wait for the food in a restaurant. It was not a pleasing experience to waste time waiting forthefood. If customers can view the menu and order the food online, they can continue doing their work while the food is prepared and delivered to their doorstep. It benefits both the restaurant as well as customers. That is because the restaurant gets more exposure online which causes an increase in sales. On the other hand, customers get their favorite foods without wasting time.

To allow customers to order food online, Sai Madhav Perumalla from the University of Bridgeport developed an online food ordering system in the fall of 2016. In that system, there were two types of users, i.e., admin and customers. The admin could manage food items whereas customers could sign in to the system, view the menu, and order food items. This system was developed to eliminate order miscommunications between customers and employees of the restaurants. It didn't allow its customers to leave feedback about the food.

In January 2019, Wong Siew Jiuan from Universiti Tunku Abdul Rahman developed a web-based restaurant ordering system. The system was expected to reduce the restaurant staff's labor work and eliminate possible inconveniences faced by the customers. In that system, customers were able to order food from the system. The system was able to forward the order status to the kitchen without any human interaction. In that system, food was delivered to the customer, and payment was collected manually by the delivery staff.

In 2011, N. A. Samsudin, S. K. Ahmad Khalid, M. F. A. Mohd Kohar, Z. Senin, and M. N.Ihkasan built a customizable wireless food ordering system with real-time customer feedback forarestaurant. It enables restaurant owners to set up the system in wireless environment and update menu presentations easily. Smartphone has been integrated into the customizable wireless food ordering system with real-time customer feedback implementation to facilitate real-time communication between restaurant owners andcustomers.

In 2015, Kirti Bhandge, Tejas Shinde, Dheeraj Ingale, Neeraj Solanki, and Reshma Totarean from Dept. of Information Technology Engineering, AISSMS, Pune, India proposed an automated food ordering system that will keep track of user orders smartly. They implemented a food ordering system for di erent types of restaurantsff in which users will make orders or make custom food with oneclick only. Employing an android application forTablet PCs, this system was implemented.

6. METHODOLOGY

6.1 SYSTEM DEVELOPMENT MODEL

In this project, we are going to use an iterative system development model. It breaks the process of developing a large application into small parts. Each part, called iteration. It represents the whole development process and contains planning, design, development, testing and maintenance. In this model functions are divided into various iterations and every iteration has its own function (task).

In first iteration, we will work on authentication and authorization with UI. First, we will analyze the requirement for authentication and design it according to requirements and planning. Then, we will go for development and testing. Finally, we will implement this iteration (authentication & authorization).

In second iteration, we will work on UI and cart. In this iteration, we will provide UI (users interface) to order food also we will provide food on the food menu to order food.
Finally, we will manage food status, and we will review the users feedback. Then we will manage menu.

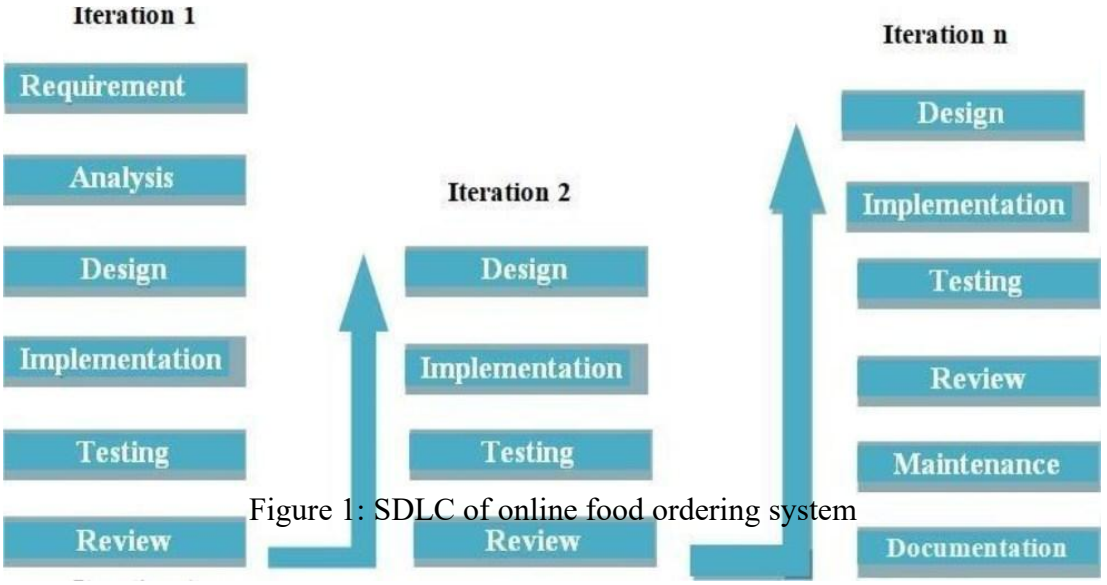


Figure 1: SDLC of online food ordering system

6.2 SOFTWARE AND HARDWARE REQUIREMENT

	Windows Requirements	Mac Requirements	Linux Requirements
Operating System	Windows7 or later	MacOsSierra10.12 or later	64 bit, Ubuntu14.04+, Debian8+, OpenSUSE13.3+, or FedoraLinux24+
Processor	IntelPentium4orlater	Intel	IntelPentium4orlater
Memory	2GBminimum,4GBrecommended		
Internet Connection	Required		

Any device capable of running a web browser is sufficient to run this system.

Frontend:

- **HTML**
- **CSS**
- **JavaScript**

Backend:

- **PHP**

Database:

- **MySQL**

Code editor:

- **Visual Studio Code**

Server:

- **XAMPP**

Web browser:

- **Chrome Browser**

6.3 USE CASE DIAGRAM

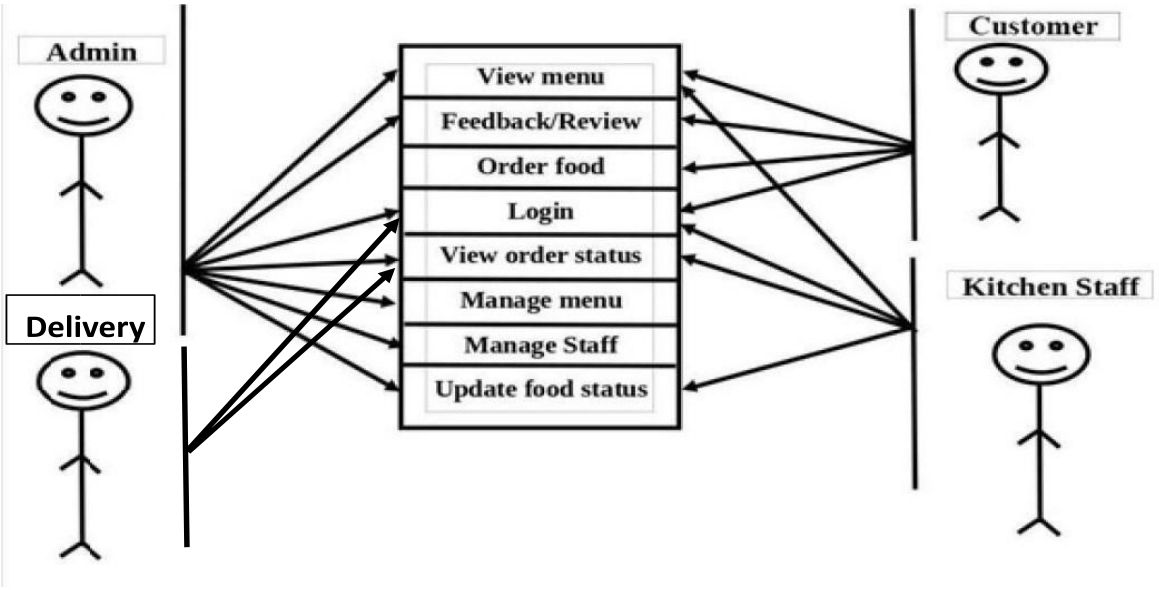


Figure 2: Use case diagram of online food ordering system.

6.4 CONTEXT DIAGRAM

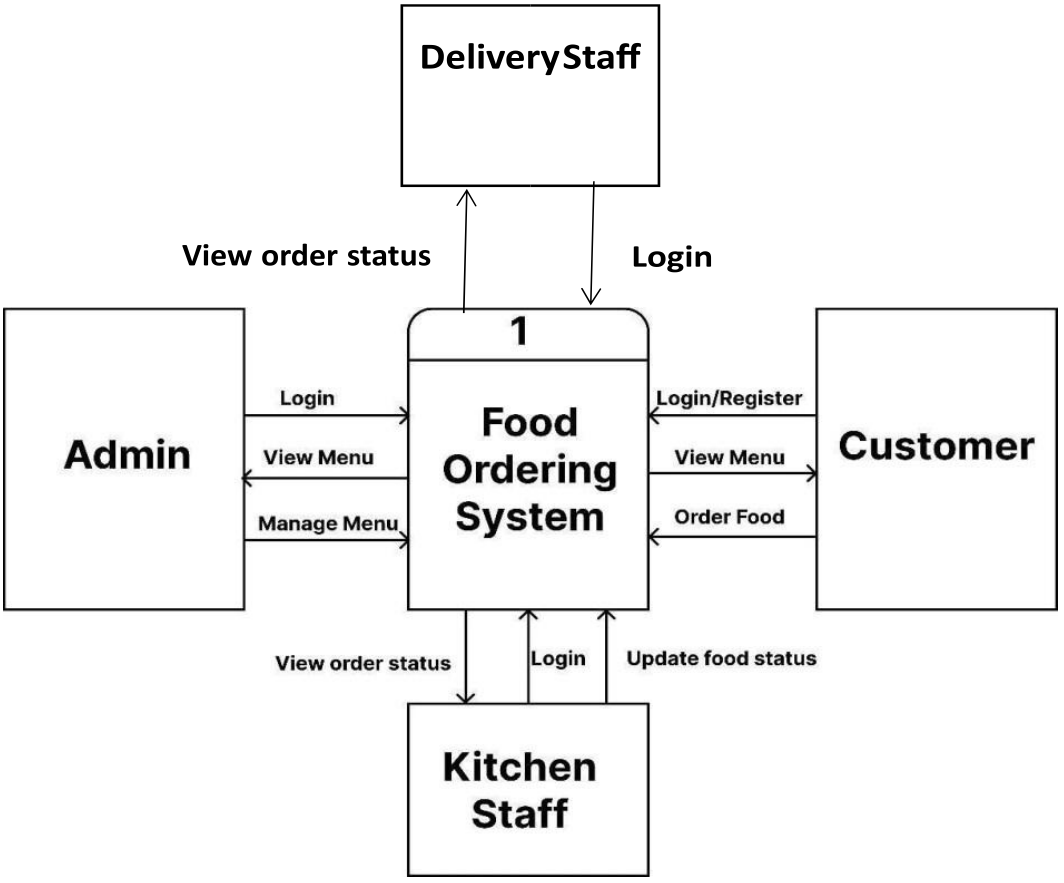


Figure 3: Context diagram of online food ordering system

6.5 DATA FLOW DIAGRAM

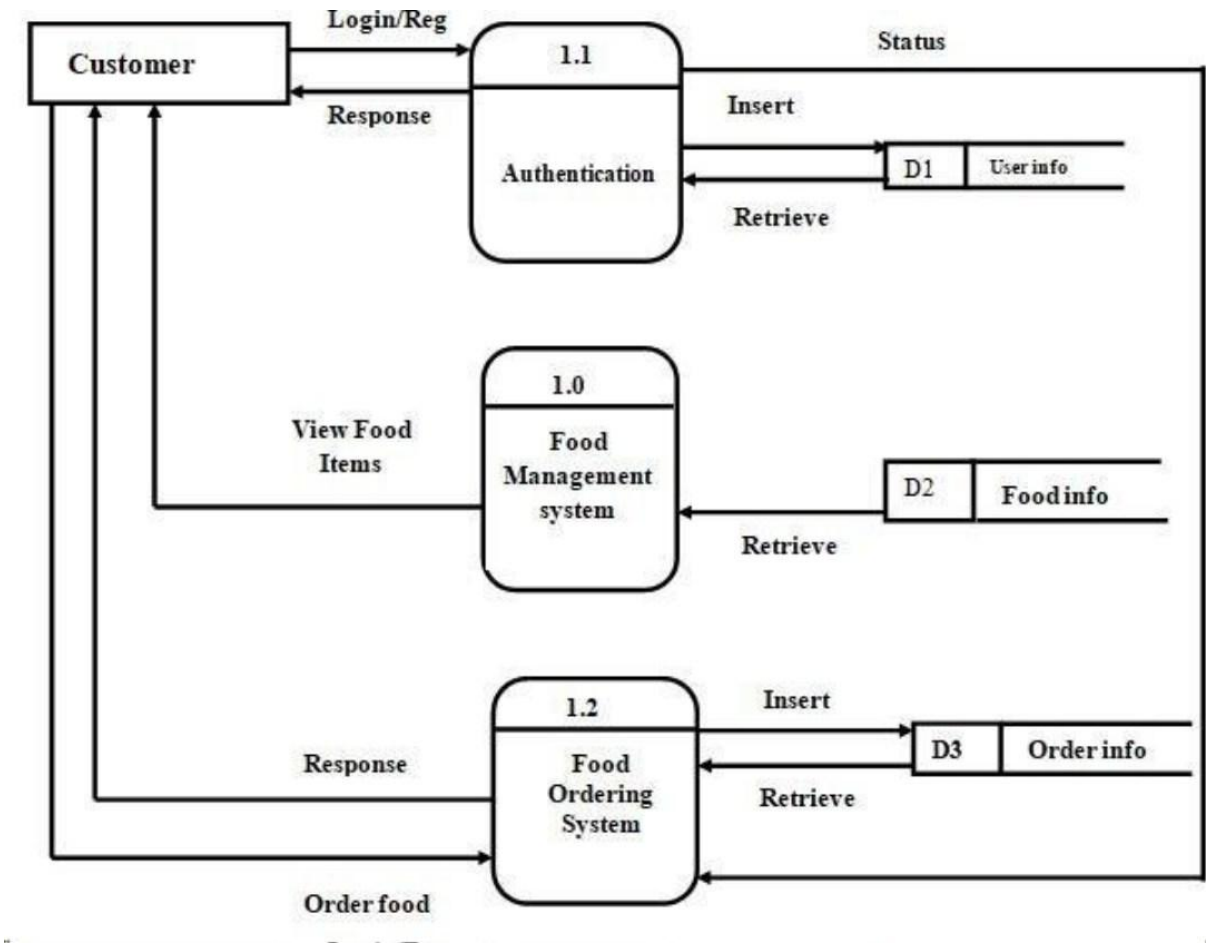


Figure 4: Customer Level 1 DFD of online food ordering system

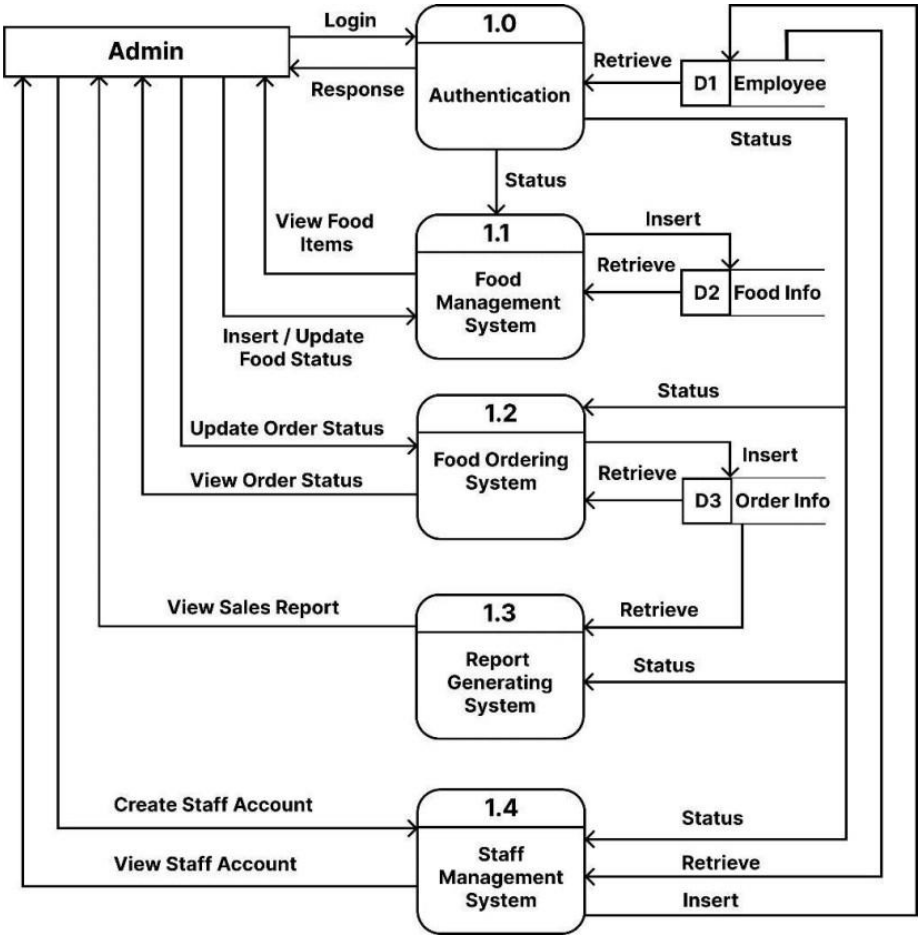
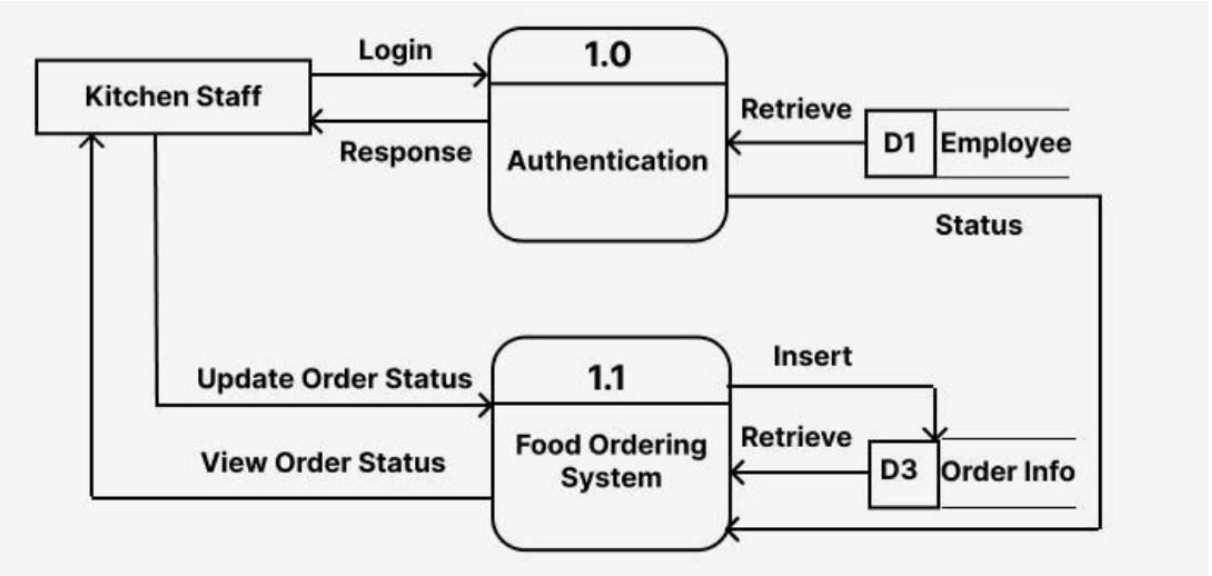


Figure 5: Admin level 1 DFD of online food ordering system



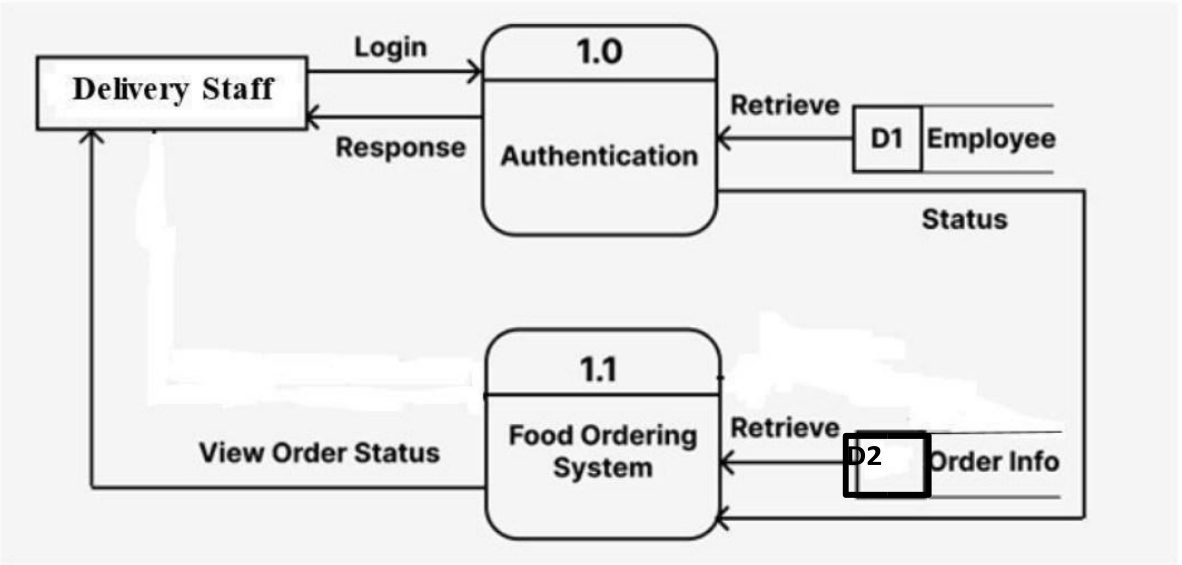
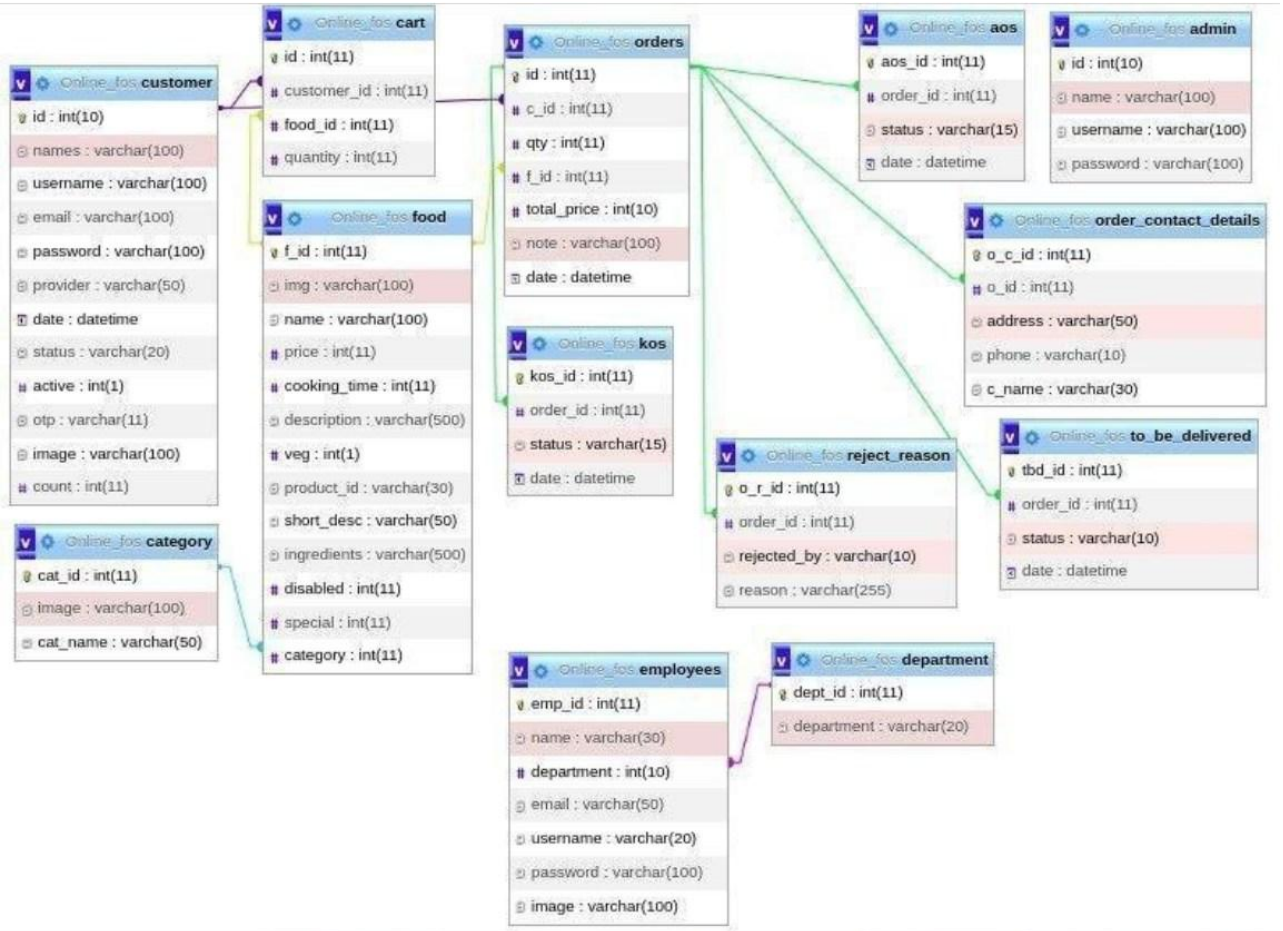


Figure 6: Kitchen Staff and Delivery staff level 1 DFD of online food ordering system

6.6 TABLE RELATIONSHIP DIAGRAM



7. RESULTS AND DISCUSSION

- Customers have the option to sign up or log in via Google.
- Customers have access to a digitalized menu where they can view all available items.
- Customers can easily place an order for their preferred food items.
- Customers can view the complete list of available food items and special dishes without logging in.
- Customers can search for their desired food items by category.
- The administrator has the authority to add or modify food items and categories.
- The administrator can add or remove special dishes from the menu.
- The administrator can activate or deactivate food items as required.
- The administrator is responsible for managing the employees.
- The administrator receives and reviews customer orders and forwards them to the kitchen.
- The administrator is authorized to accept or reject customer orders.
- The kitchen has access to all ordered items which is accepted by administrator.
- The delivery has access to the list of prepared foods.

Customers can conveniently register or log in using their Google or Facebook accounts. Once logged in, customers can access a digital menu to view all available items and easily place an order for their favorite food items. Additionally, customers can view a comprehensive list of available and special dishes without the need to log in and can search for specific food items by category.

The administrator is responsible for managing the website's food items, categories, and employees. They have the authority to add or edit food items and categories, add or remove special dishes from the menu, and enable or disable food items as needed. Furthermore, the administrator receives and reviews customer orders, forwarding them to the kitchen. They also can accept or reject customer orders as needed.

The kitchen has access to all ordered items and can view the complete list of food ordered items. Overall, this system streamlines the ordering process for customers while providing effective management tools for the administrator and kitchen staff.

The delivery has access to the list of prepared foods. Overall, this system streamlines the ordering process for customers while providing effective management tools for the administrator and delivery staff.

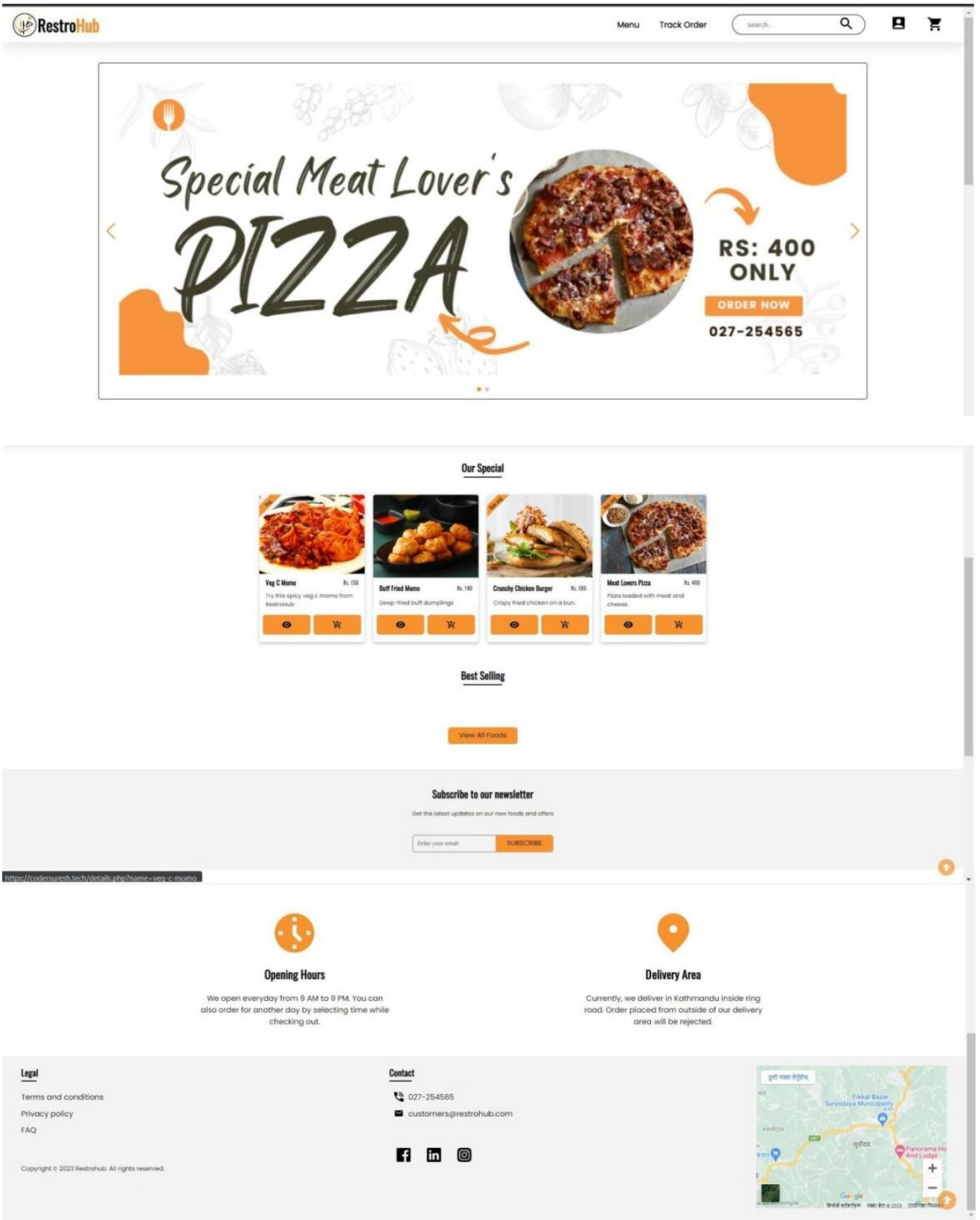


Figure 8: index page

This is the first page of our system. Here customers can view special items and menu without logging in.

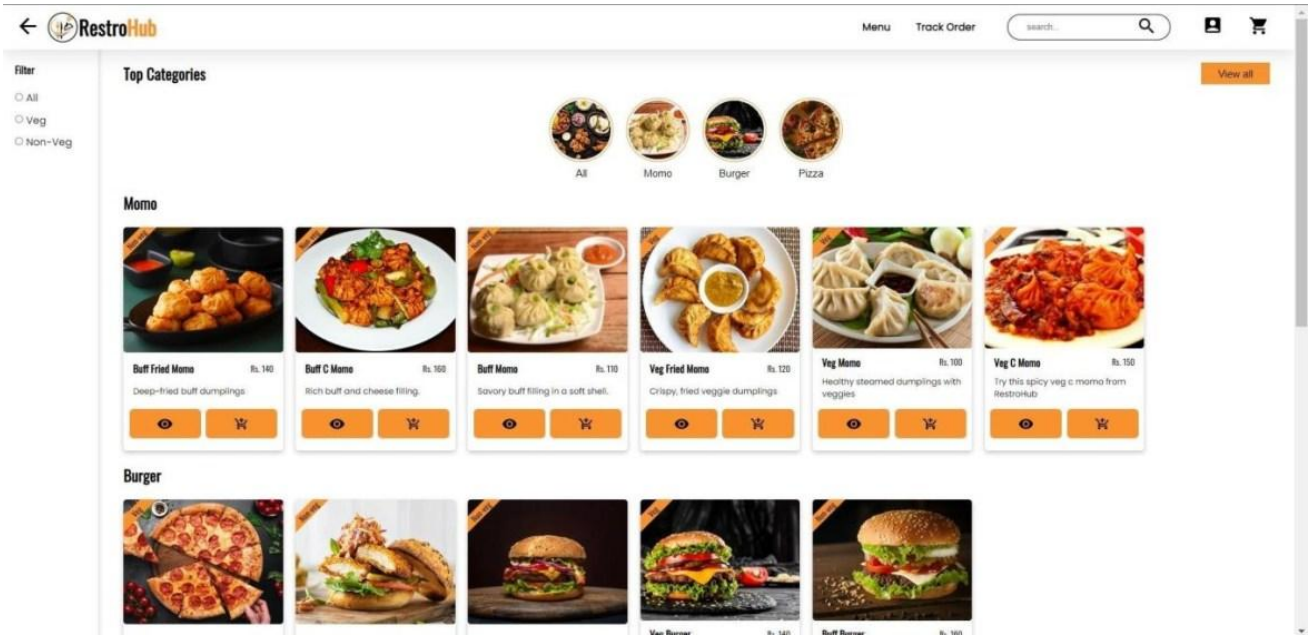


Figure 9: menu

In this page customer can view all the available item in the menu without logging in.

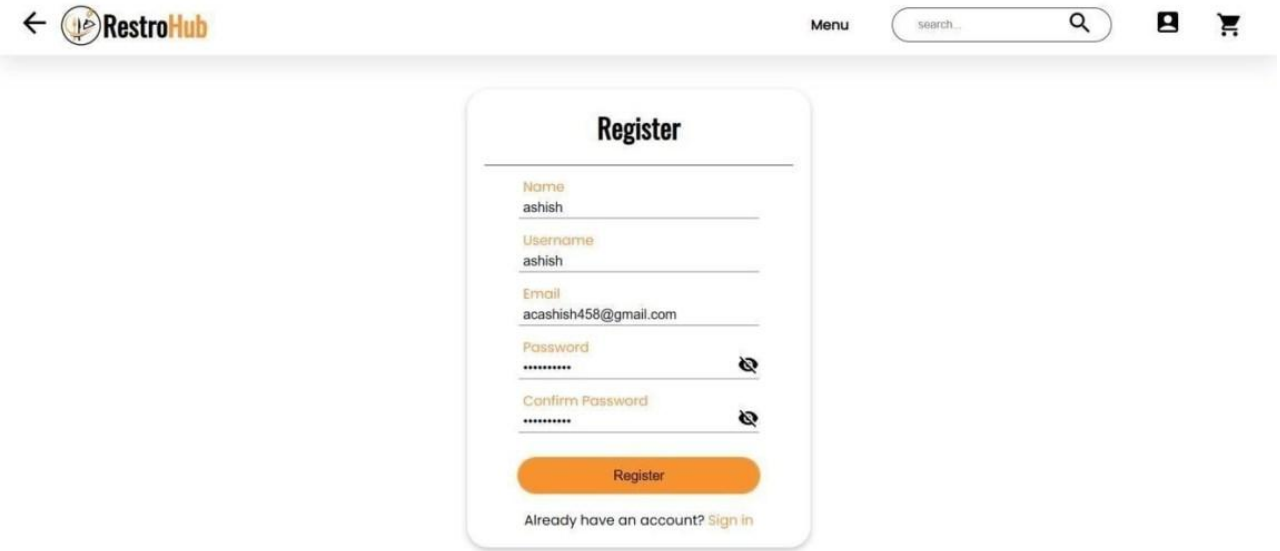


Figure 10 register page

This page is used to register a new customer. After successful registration, customer will be redirected to login.php. If the registration is failed it gives error message.

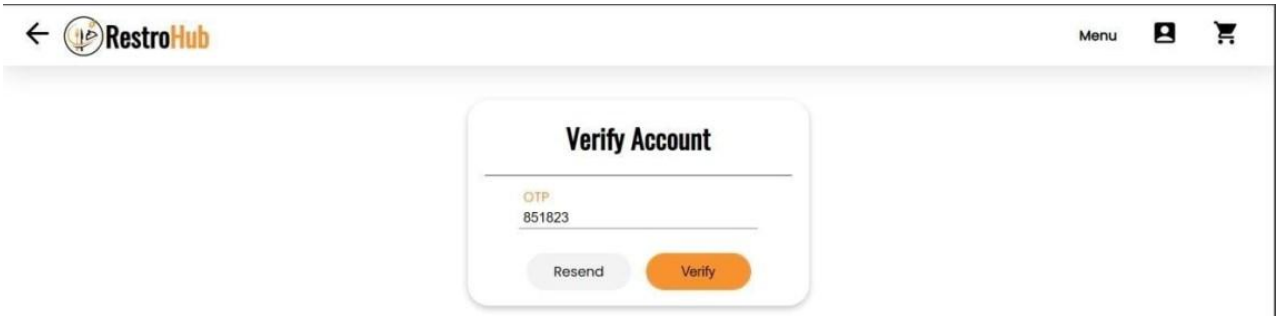


Figure 11: verification page

To verify the customer, customers should provide valid email addresses, and our system will send otp code to verify the customer.

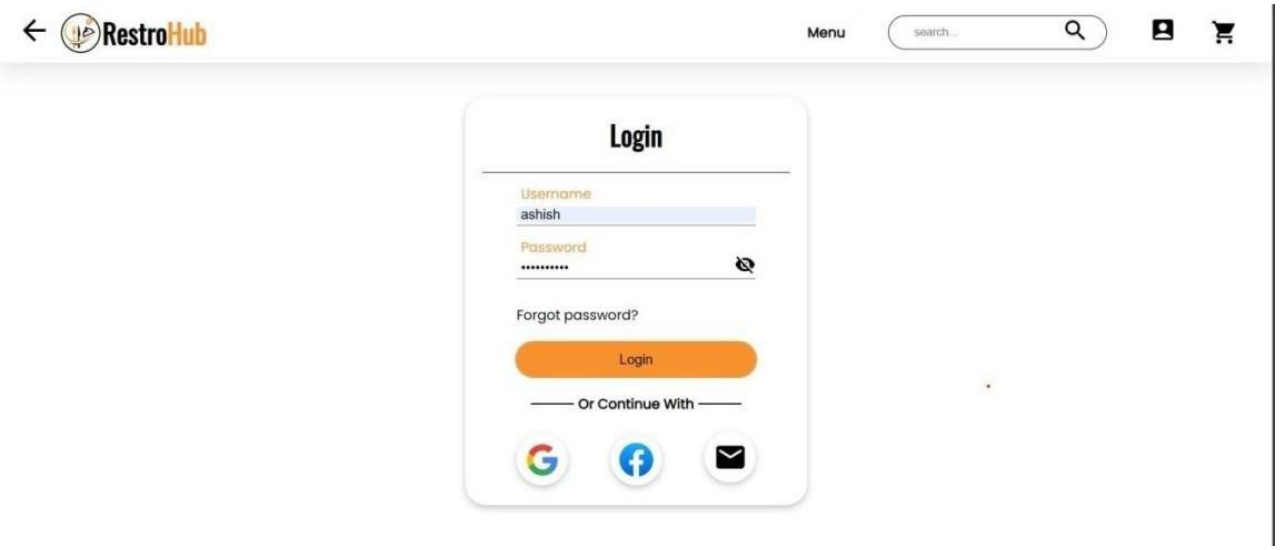


Figure 12: login page

After successful registration, customers can login in the system through this page.
Customer can directly login through them google account also.

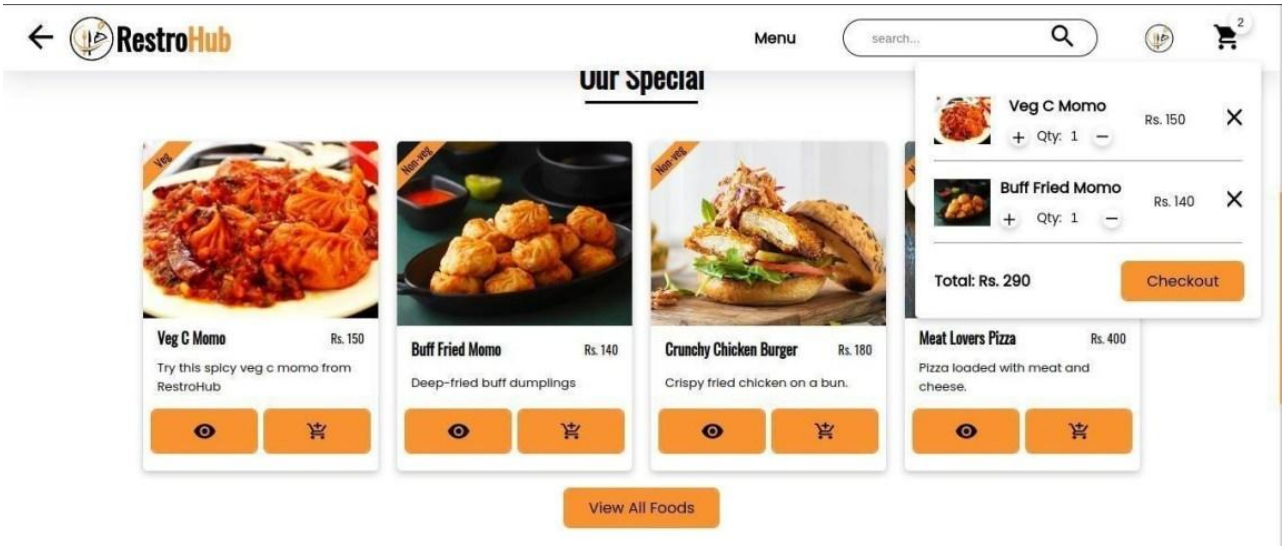


Figure 13: index page (cart)

Customer can add multiple items to cart and see those items.

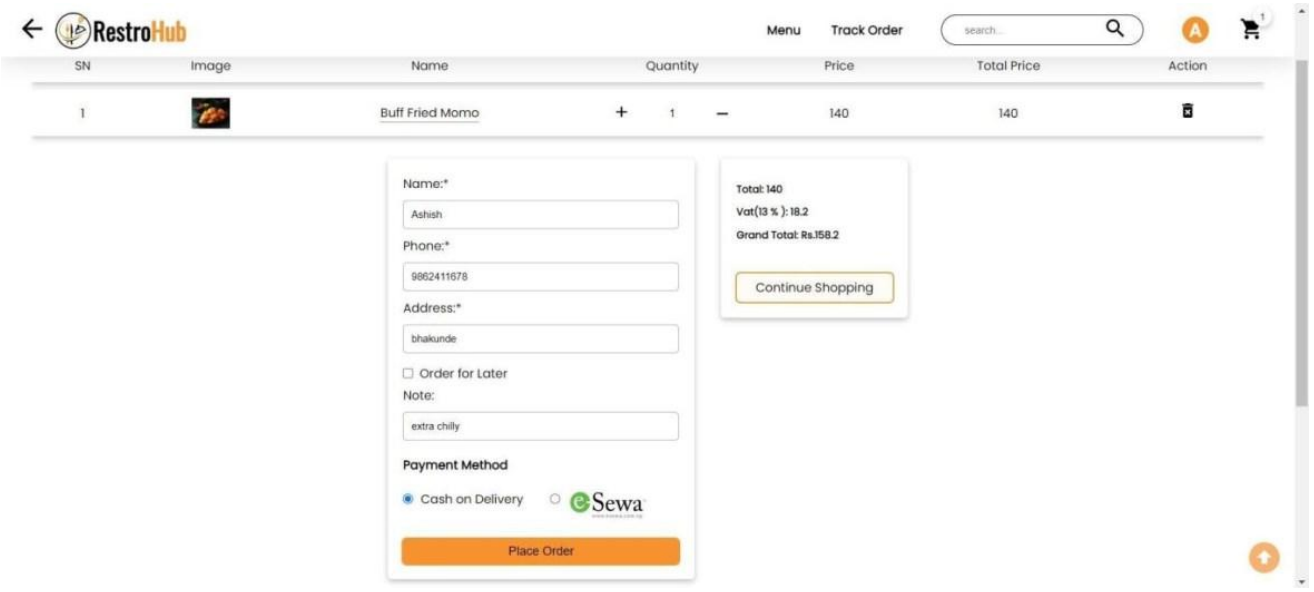


Figure 14: Checkout for quick order (cash on delivery payment)

After adding items to cart, customer can place an order by providing necessary details through this page.

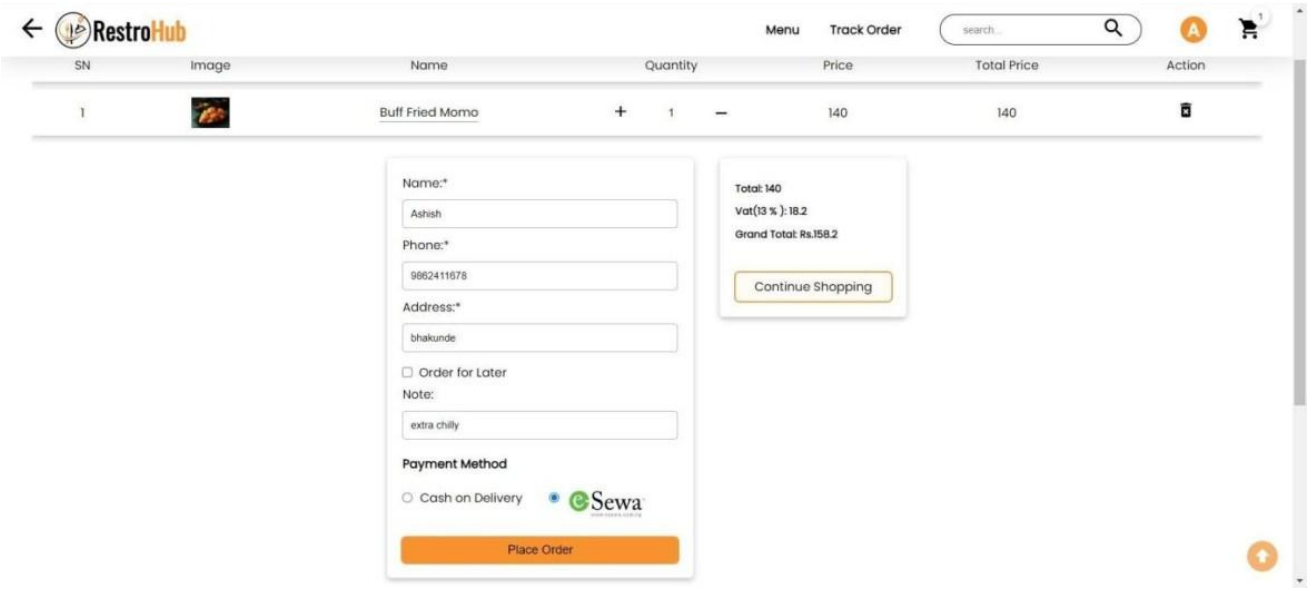


Figure 15: checkout for quick order (E-sewa payment)

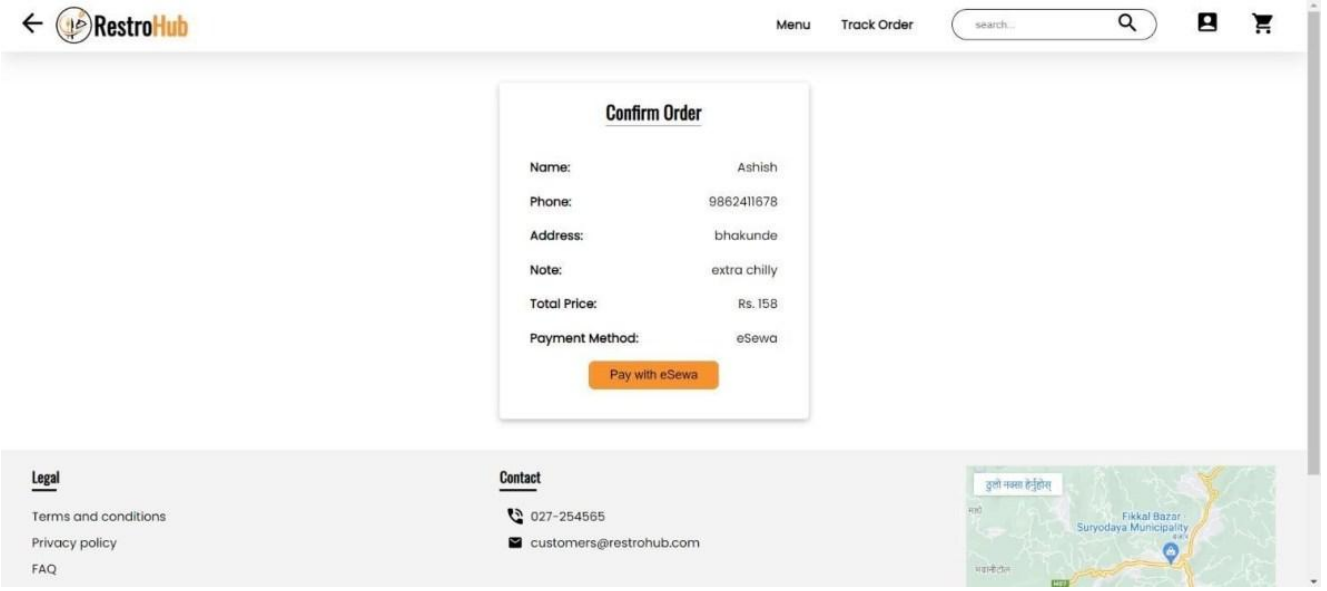


Figure 16: Conform order to e-sewa

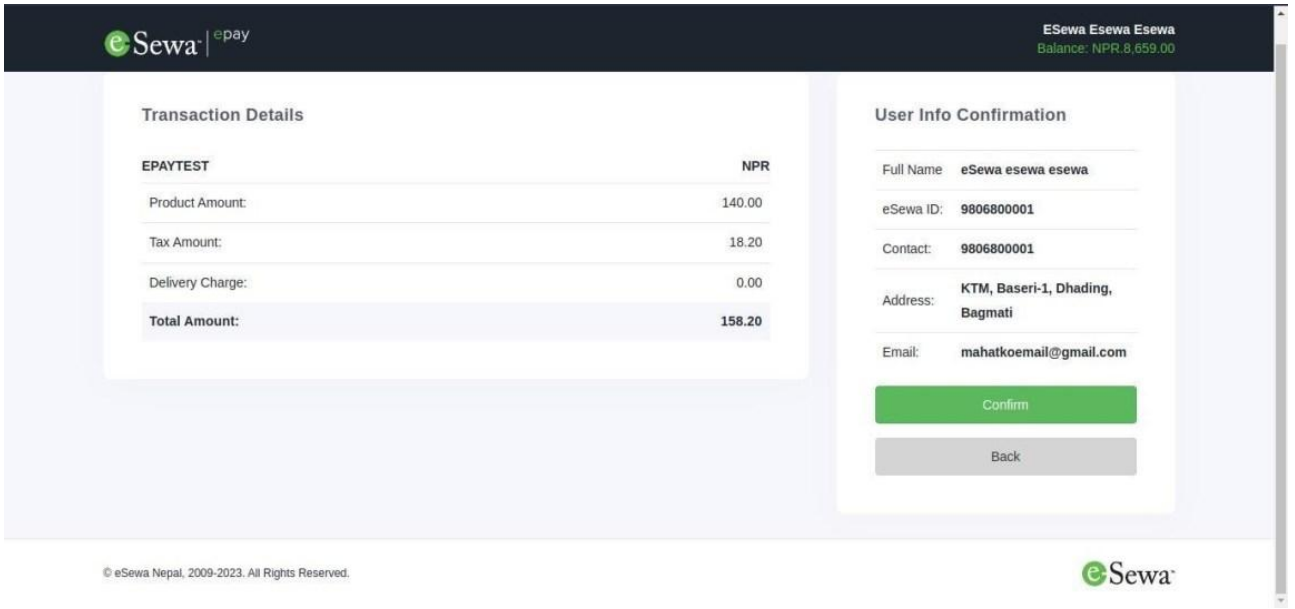


Figure 17: E-sewa payment confirm.

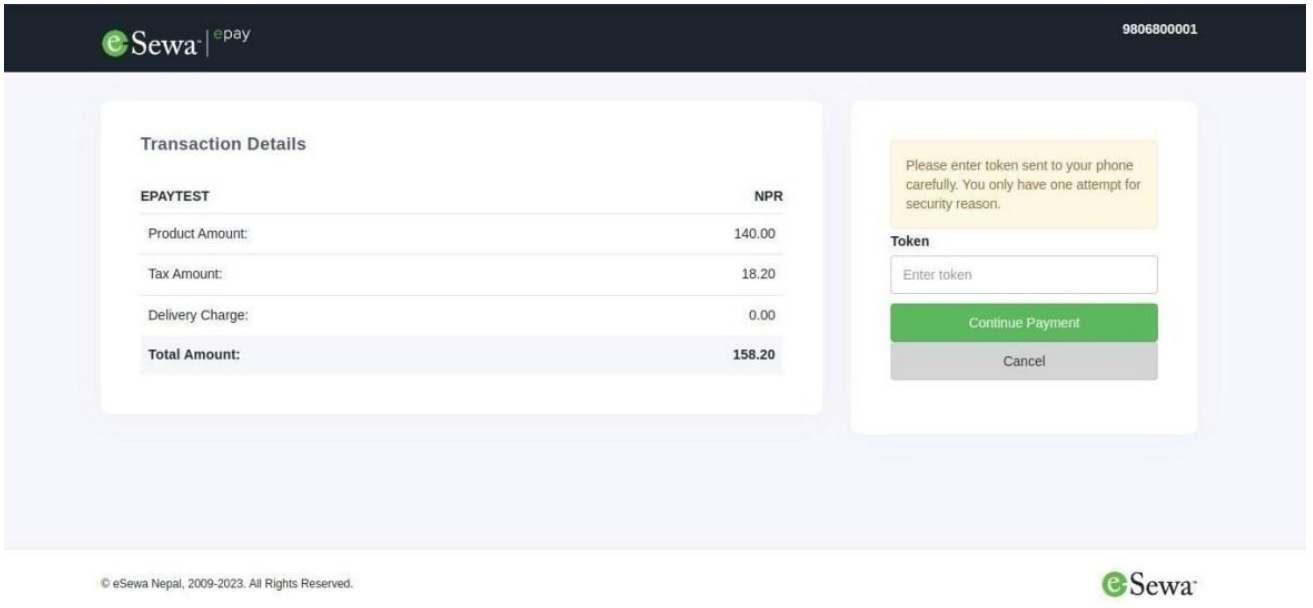


Figure 18 Token through e-sewa

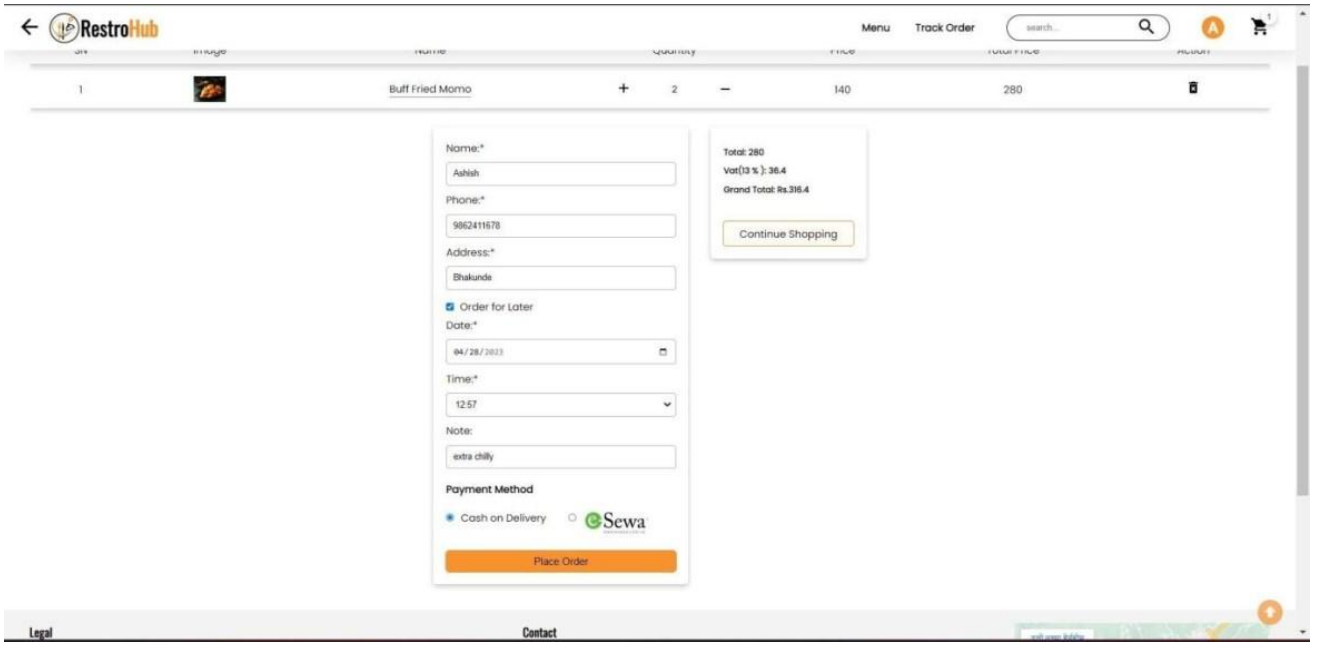


Figure 19: Checkout of order for later (cash on delivery payment)

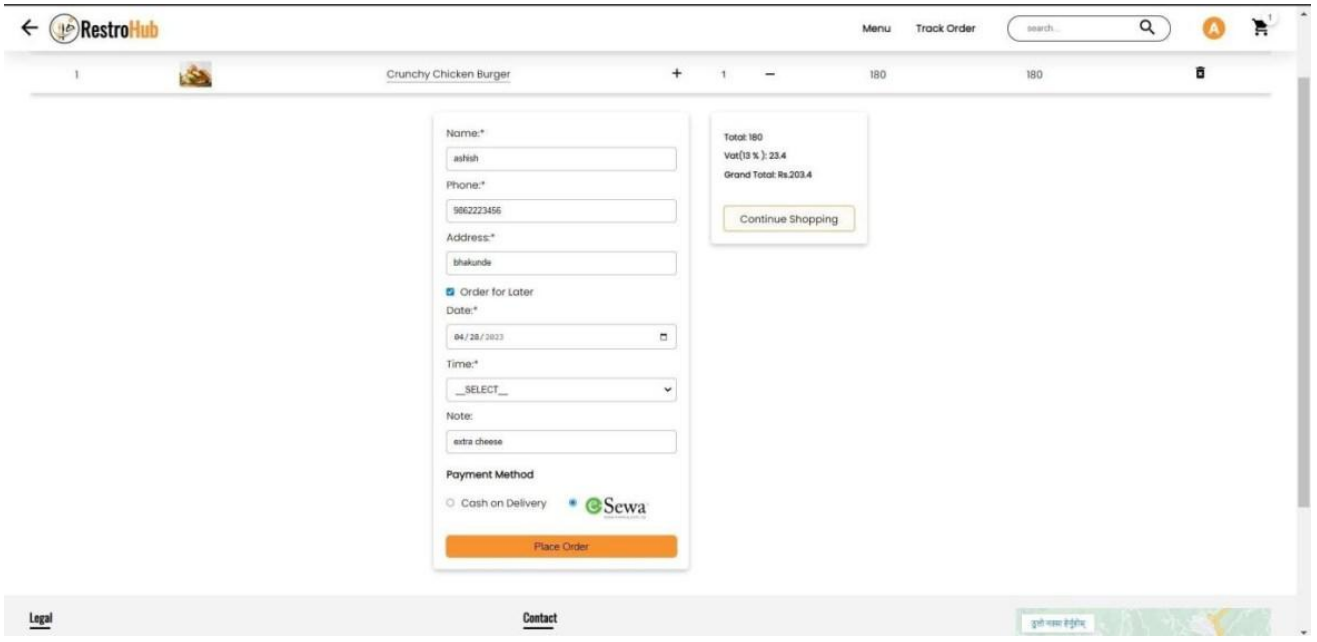


Figure 20: Checkout of order for later (E-sewa payment)

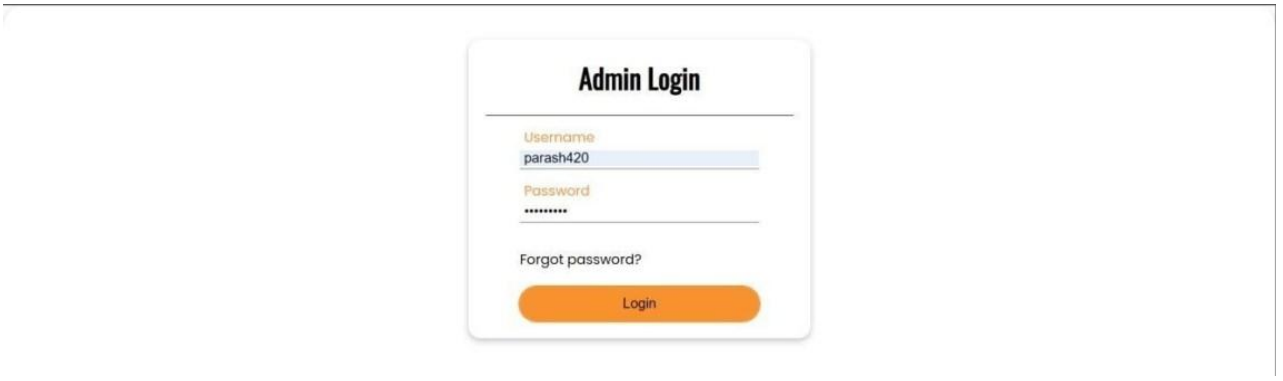


Figure 21: login page (admin)

Admin can login in the system through this page. Admin should provide valid information to login successfully.

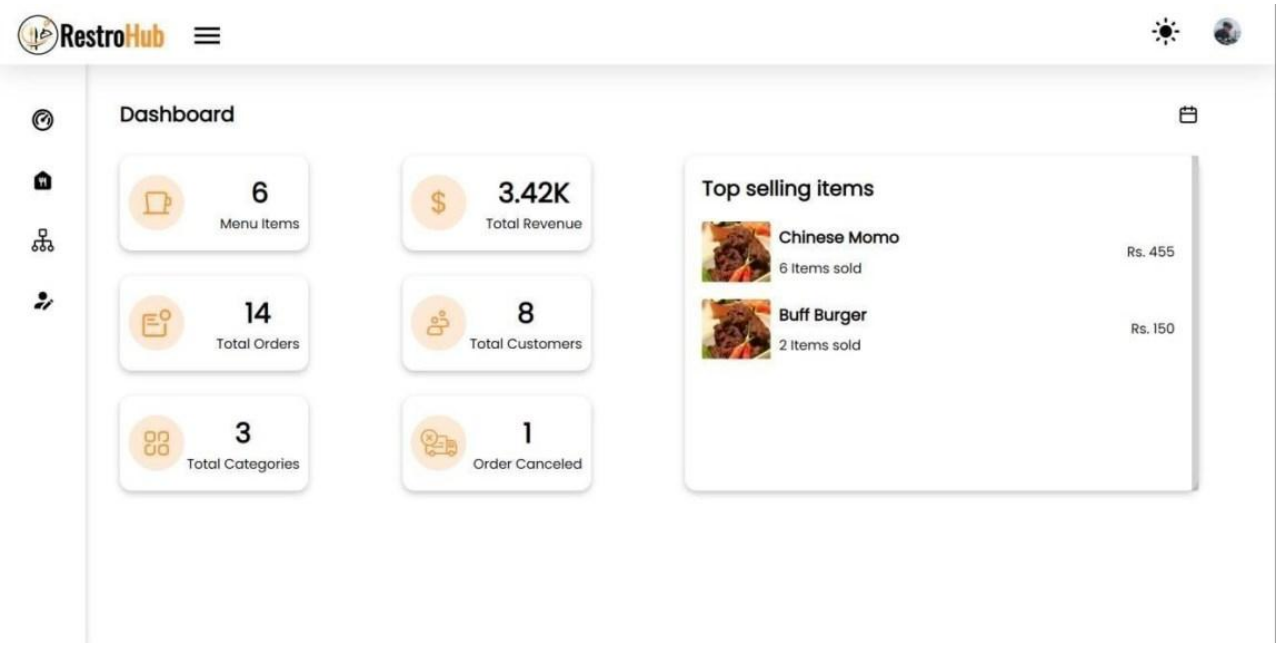


Figure 22: Index page (admin)

After successful login admin in redirected to this page. This is the dashboard for admin. Admin can see all the details of the system as well as manage employees, foods, categories, and orders.

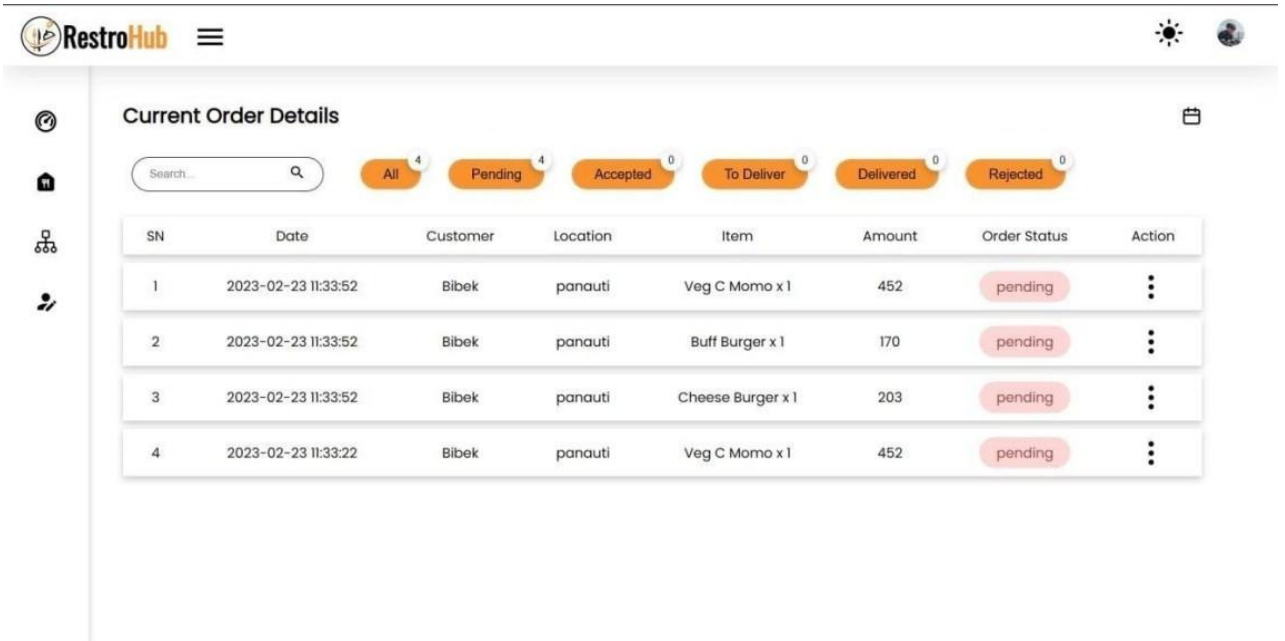


Figure 23: Order detail page

This page shows all the current order details. Admin can accept or reject the order and forward order to the kitchen. This page also provides the status of order.

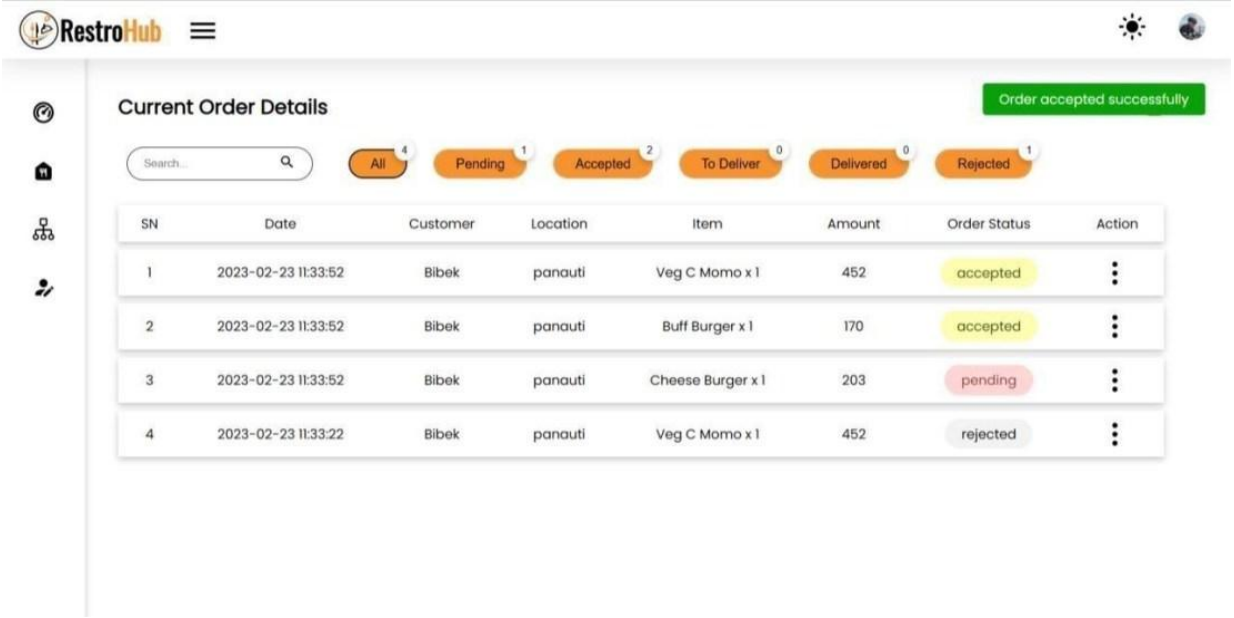


Figure 24: Order detail page

After accepting the order of customer, order status is changed to accept.

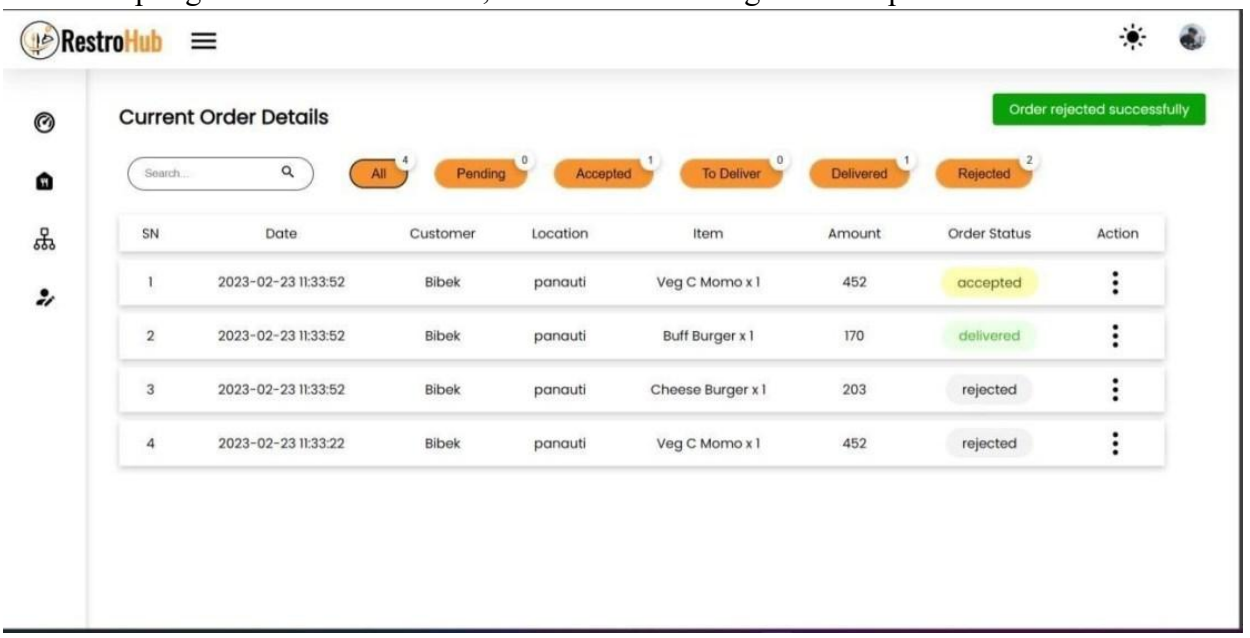


Figure 25: Order detail page

When admin reject the order, order status is changed to reject.

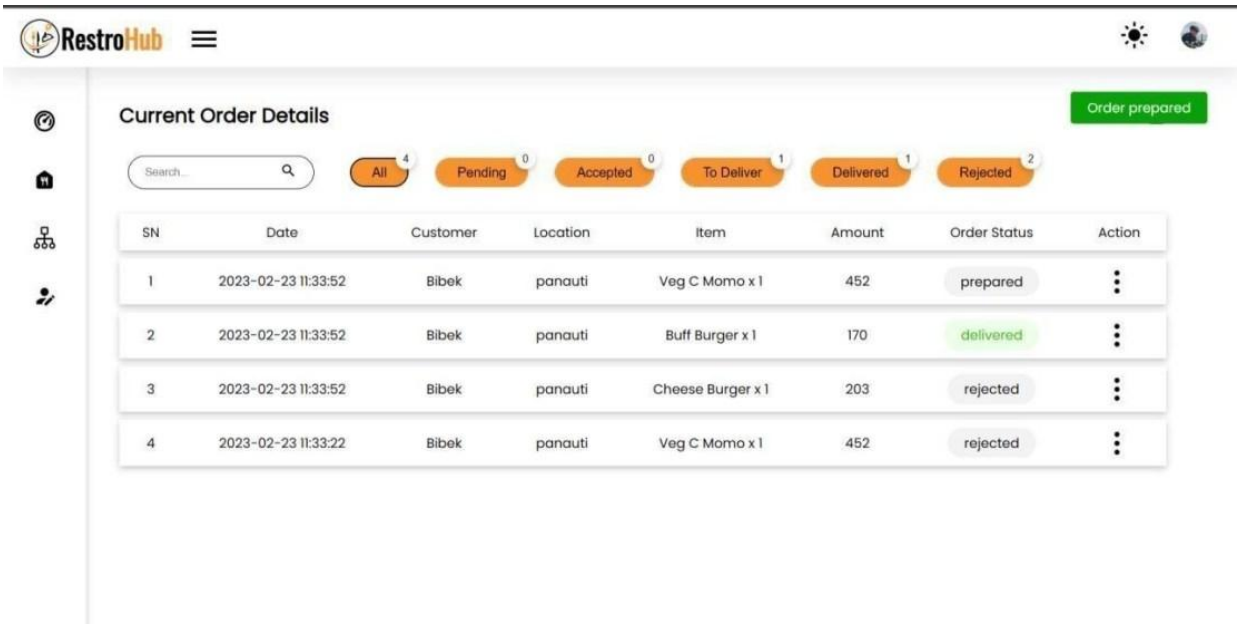


Figure 26: Order detail page

When an order is prepared successfully, the order status is changed to prepare.

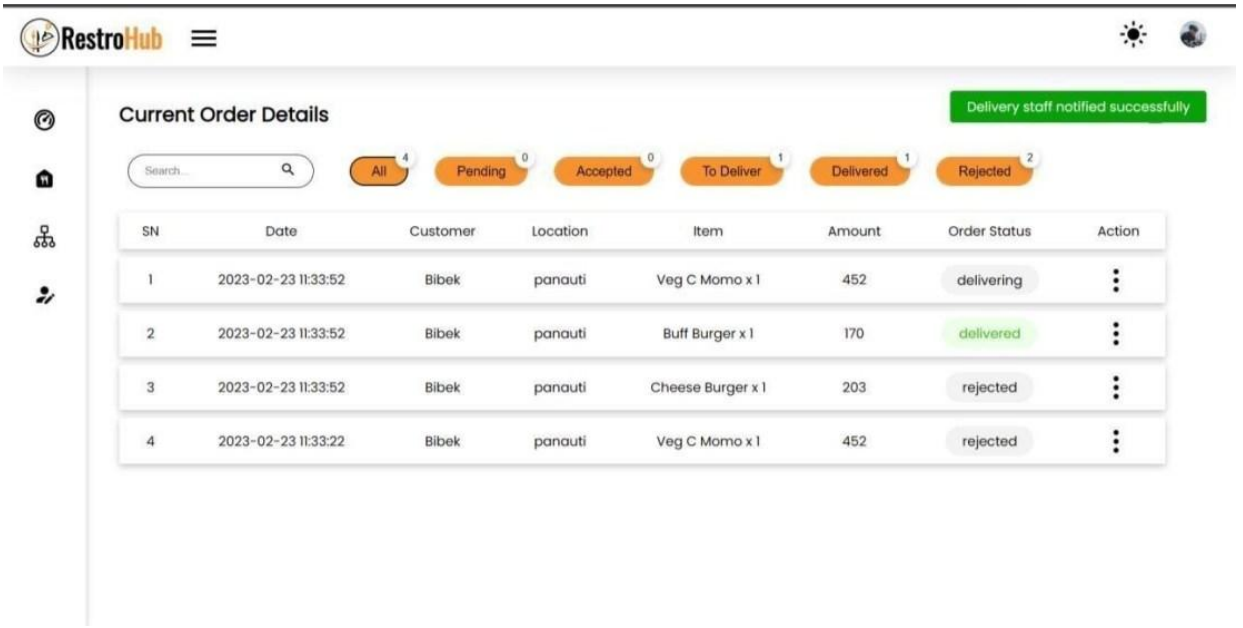


Figure 27: Order detail page

When order is being delivered by delivery boy, the order status is changed to “delivering”.

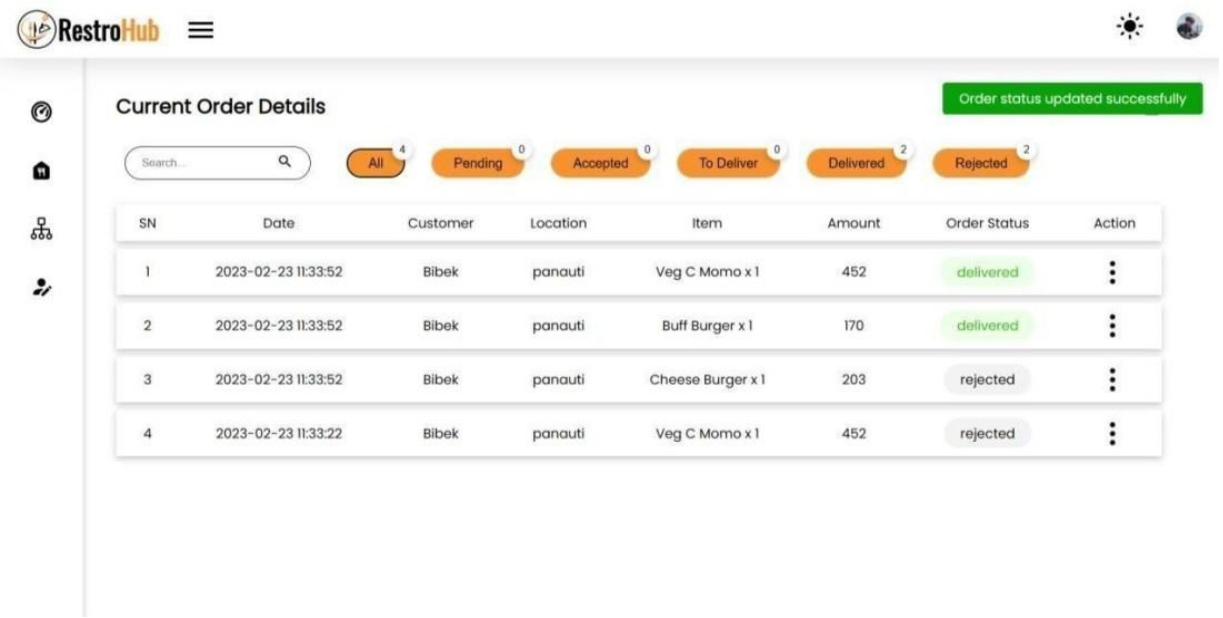


Figure 28: Order detail page

After successful delivery the order status is changed to “delivered”.

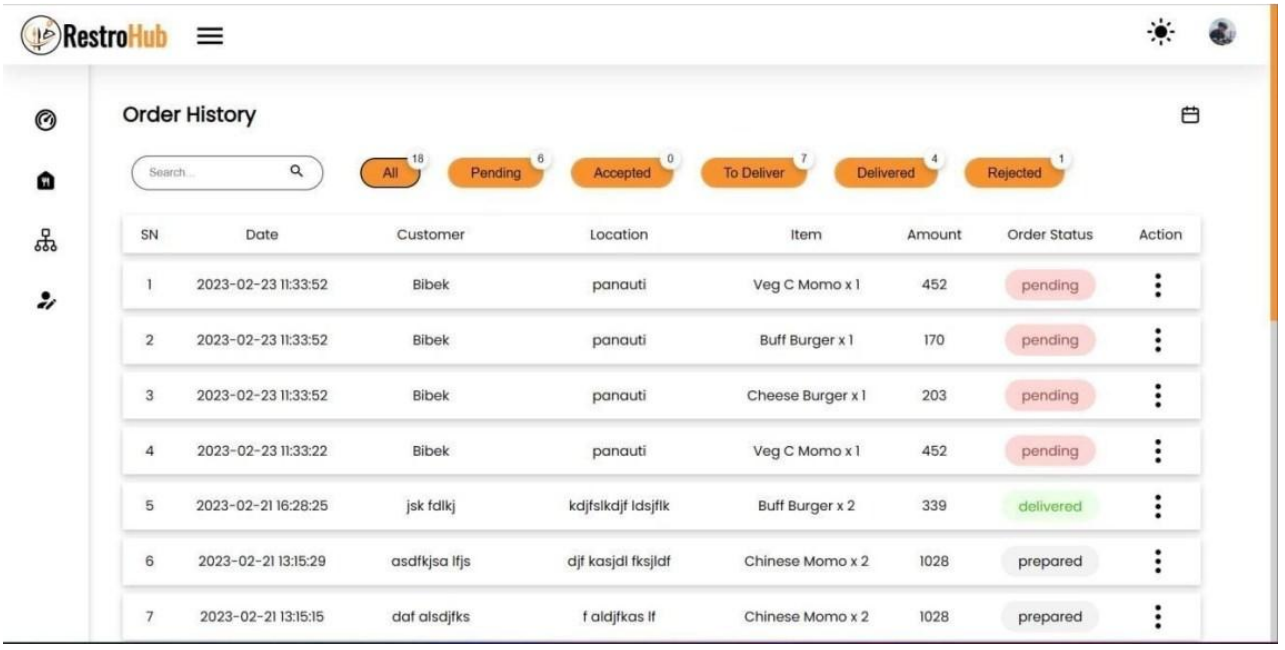


Figure 29: Order detail page

This section shows order history made by customers and all the other necessary details.

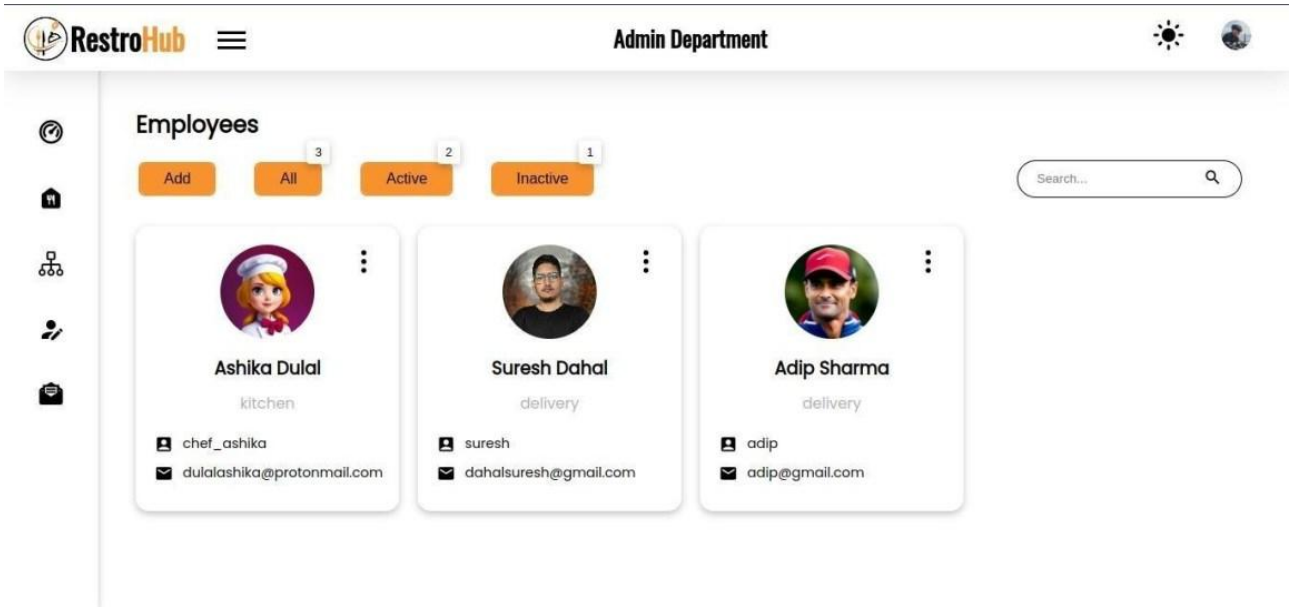


Figure 30: Employee management page (admin)

Admin can add or remove employees from this page.

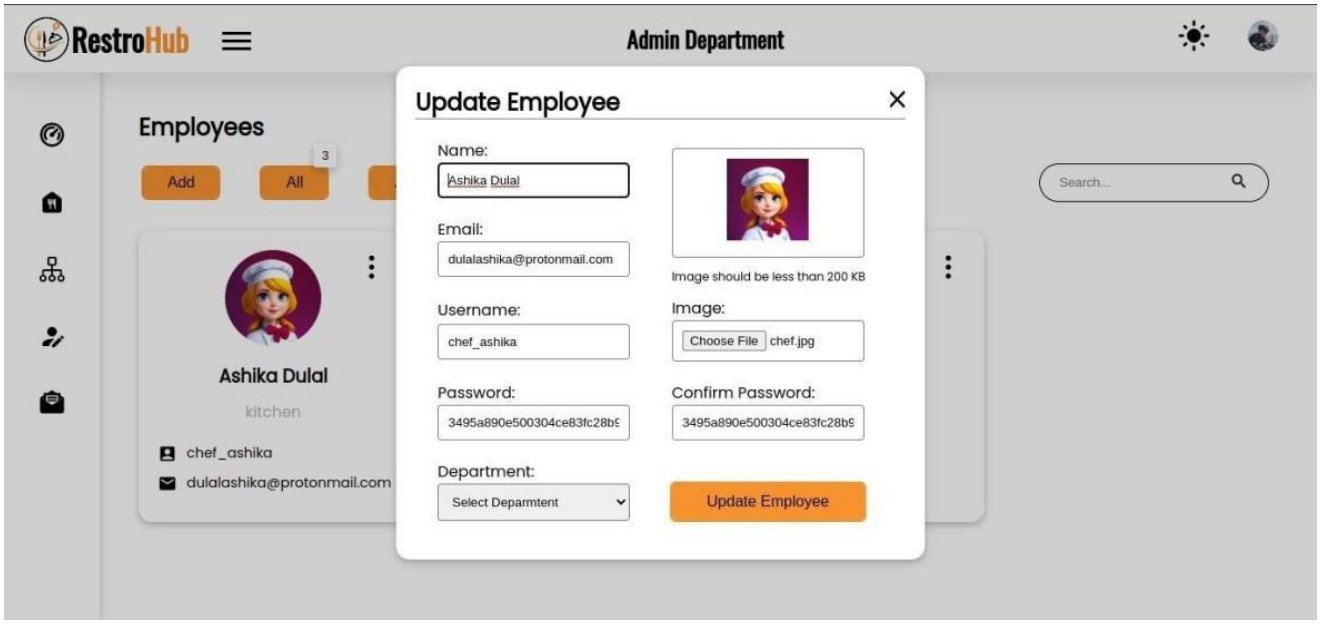


Figure 31: Employee management page (admin)

Admin can update employee details from this page.

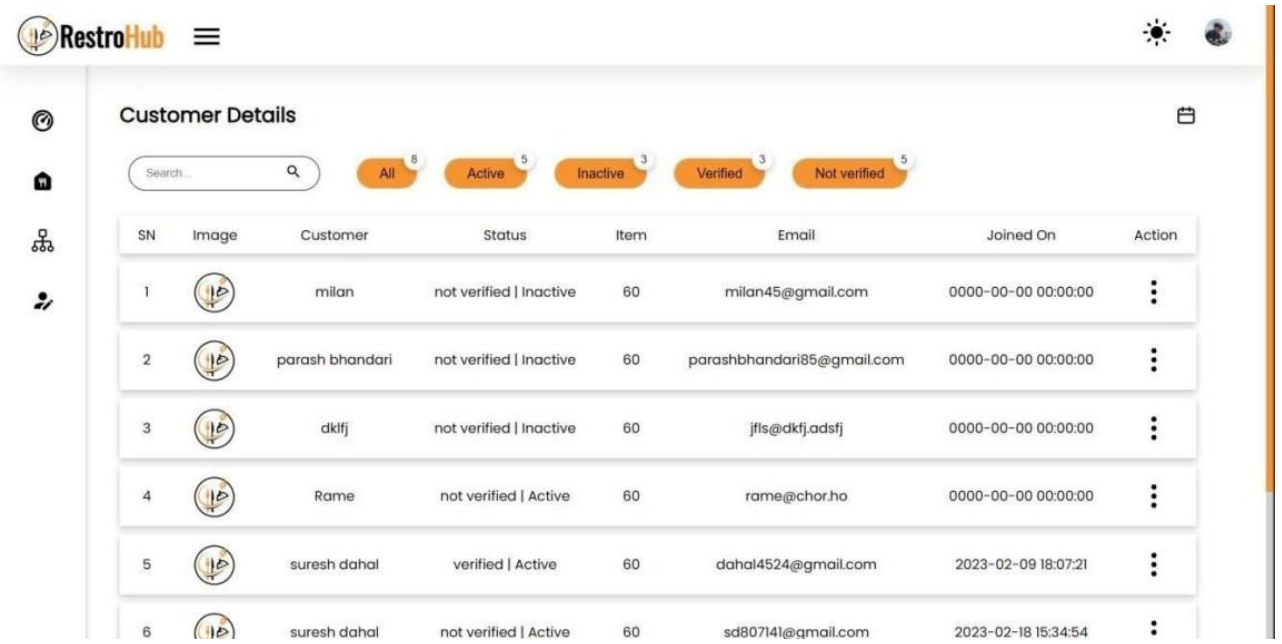


Figure 32: Customer detail page.

This page provides all the necessary details of customers.

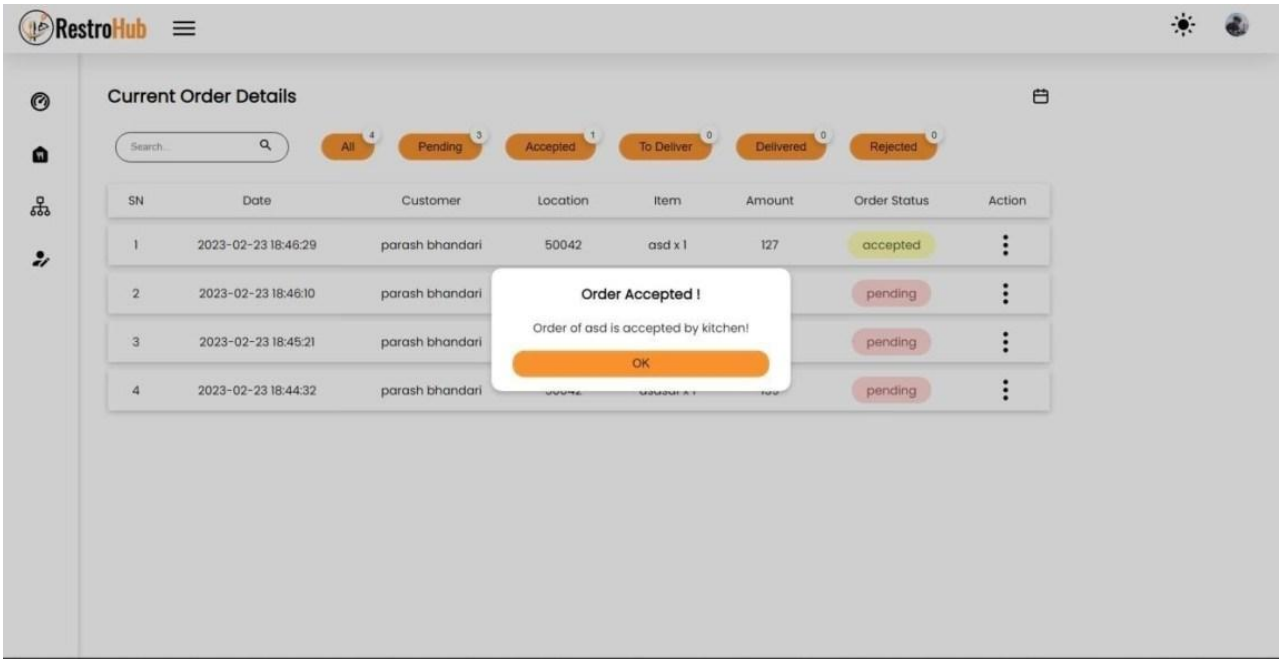


Figure 33: Order detail page.

When a customer makes a new order, a notification is shown in admin section.

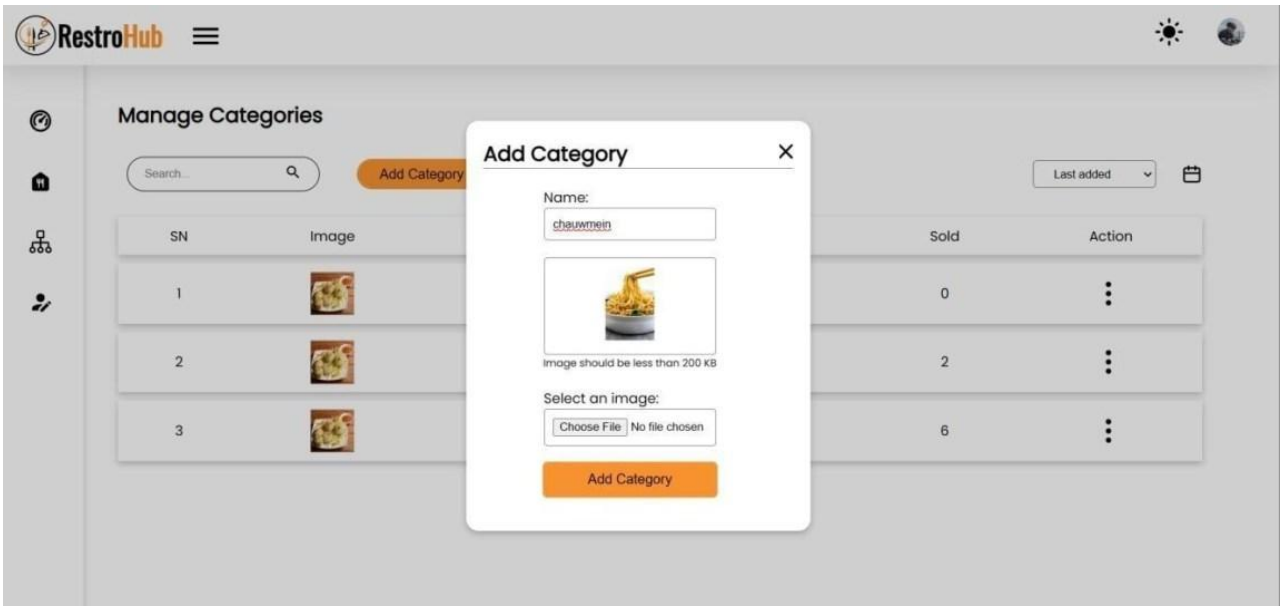


Figure 34: Add food page.

From this page admin can add new food item to the menu by providing necessary details

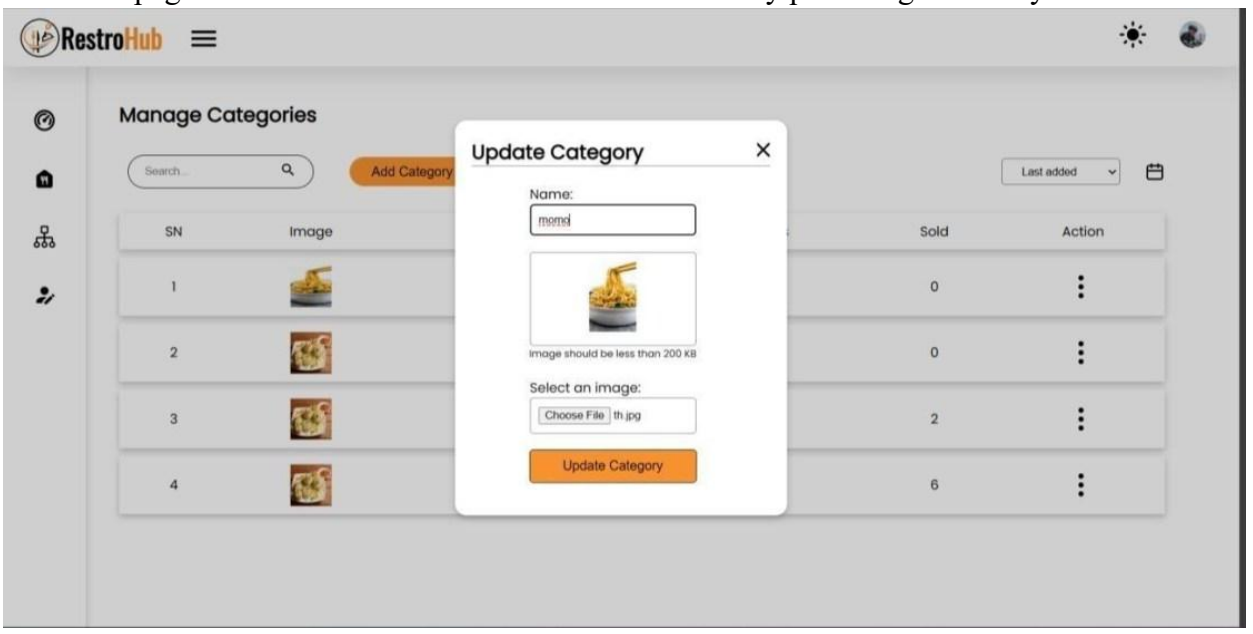


Figure 35: Update categories page.

From this page admin can edit the existed categories.

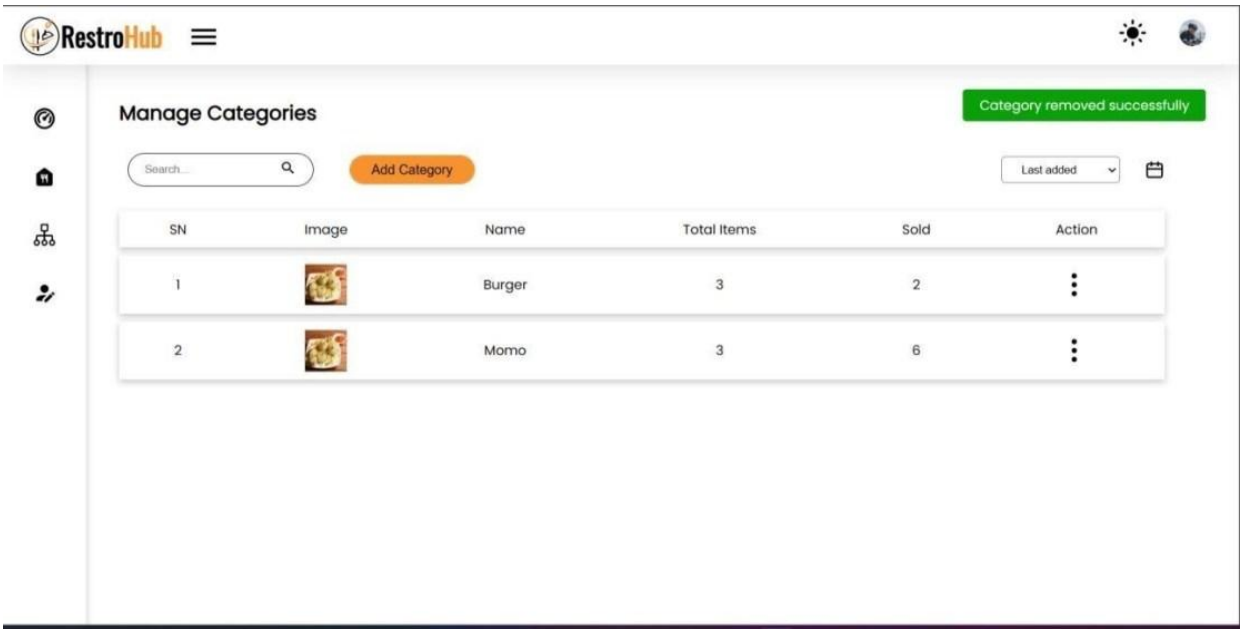


Figure 36: Categories management page.

From this page admin can remove empty food category.

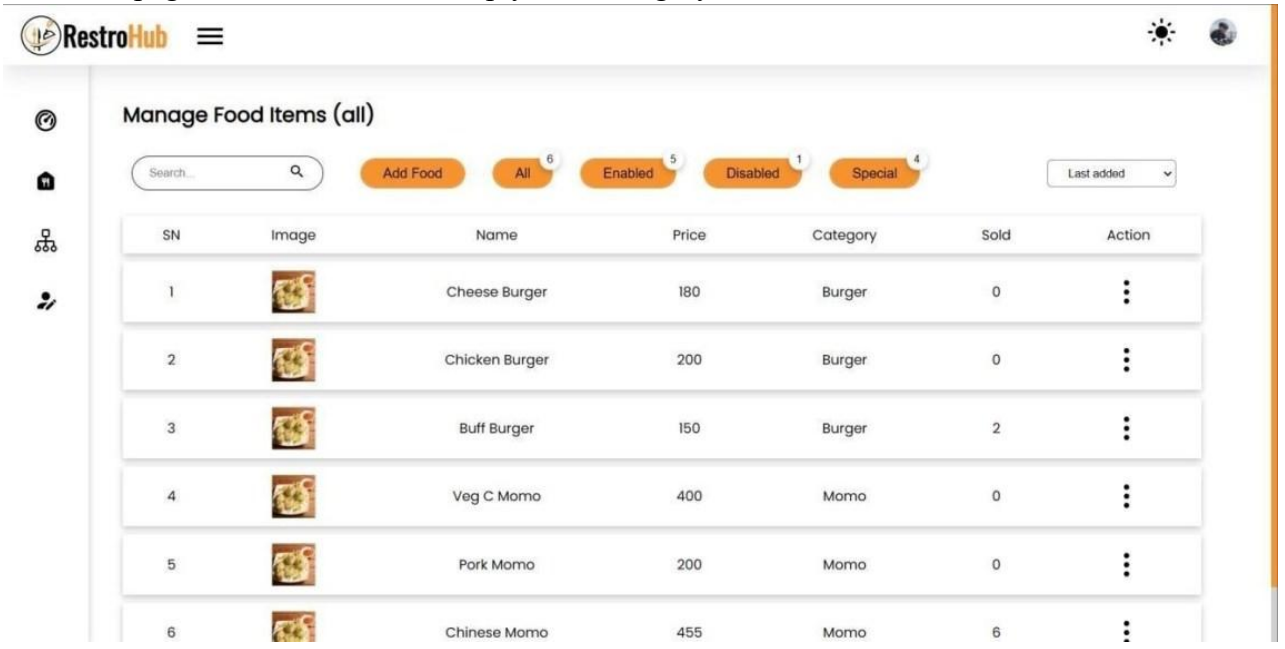


Figure 37: Food management page

From this page admin can manage all the food items. Admin can enable, disable food items as well as admin can remove or add items from special section.

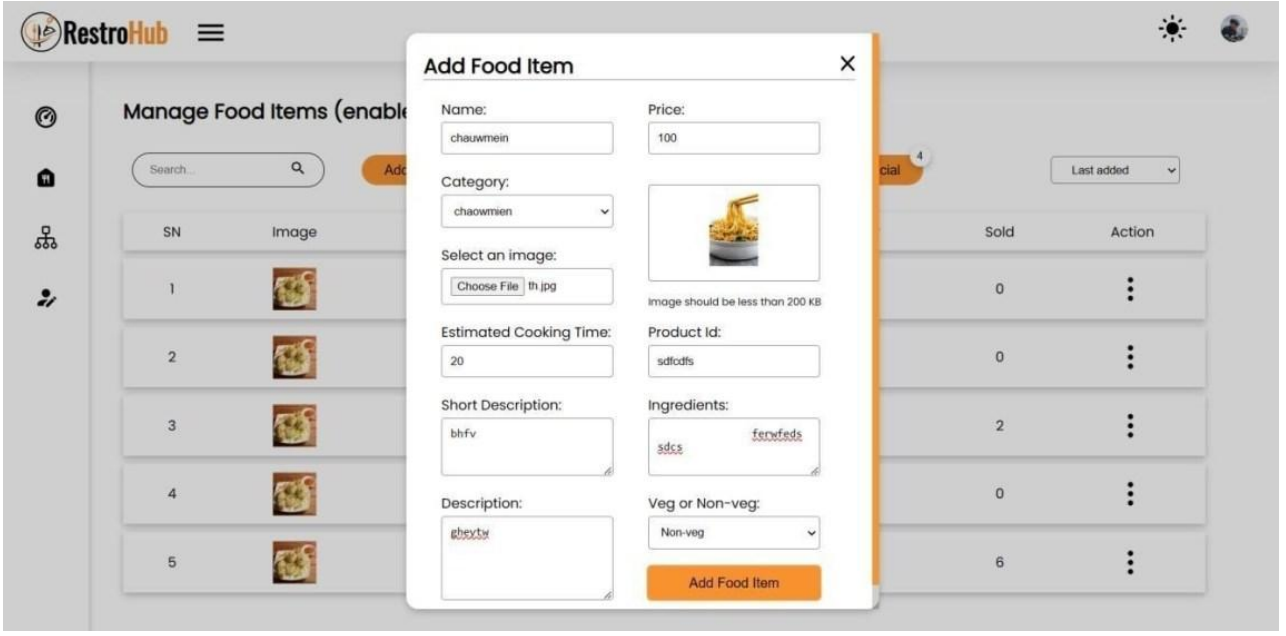


Figure 38: Food item adding page

Admin can add new food item to the menu by providing necessary details and image of the food.

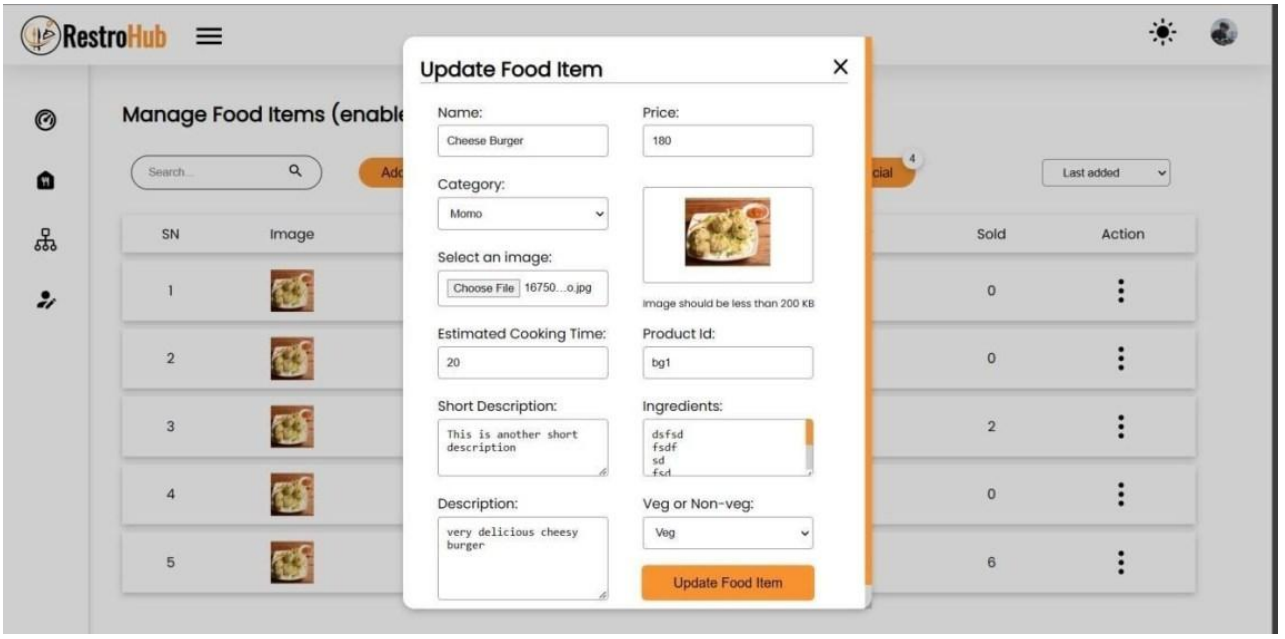


Figure 39: Food item updating page

From this page admin can edit existing food item details.

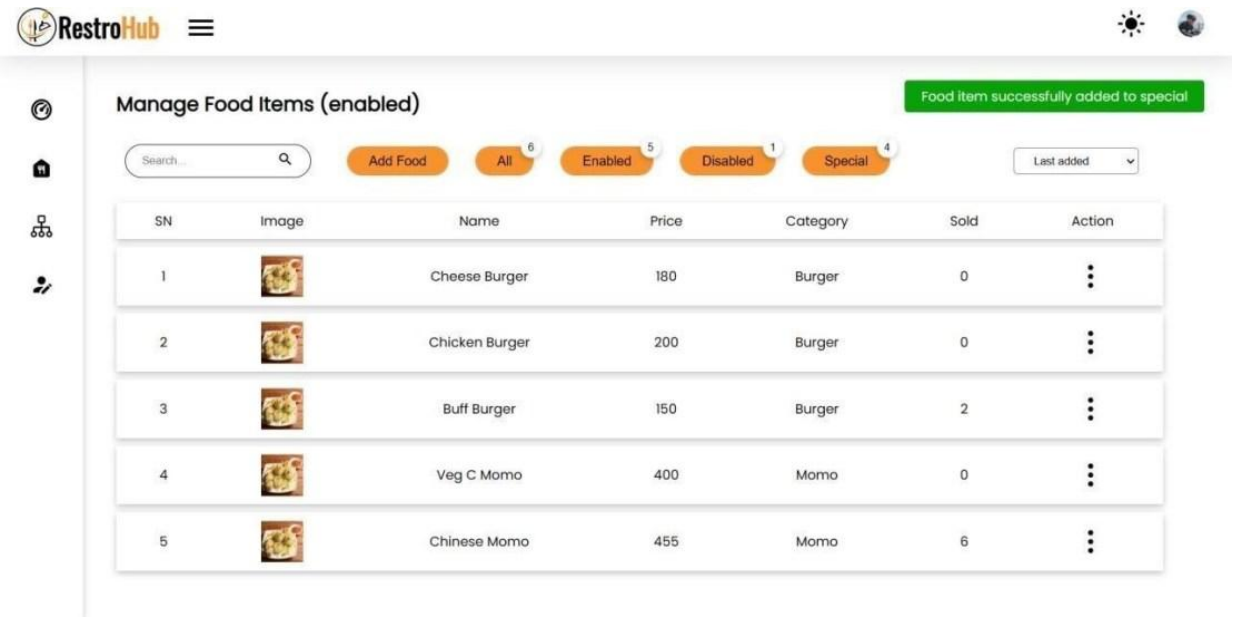


Figure 40: Food management page

When admin add food items to the special section it gives an message as shown in above figure.

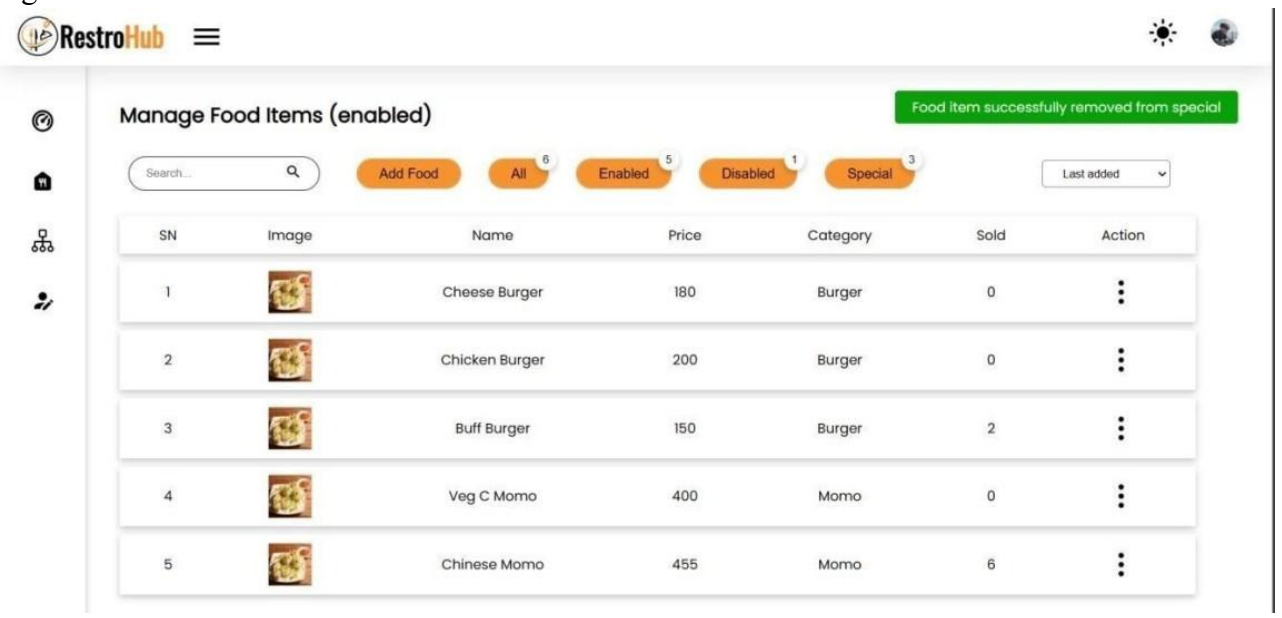


Figure 41: Food management page

When admin remove food items to the special section it gives an message as shown in above figure.

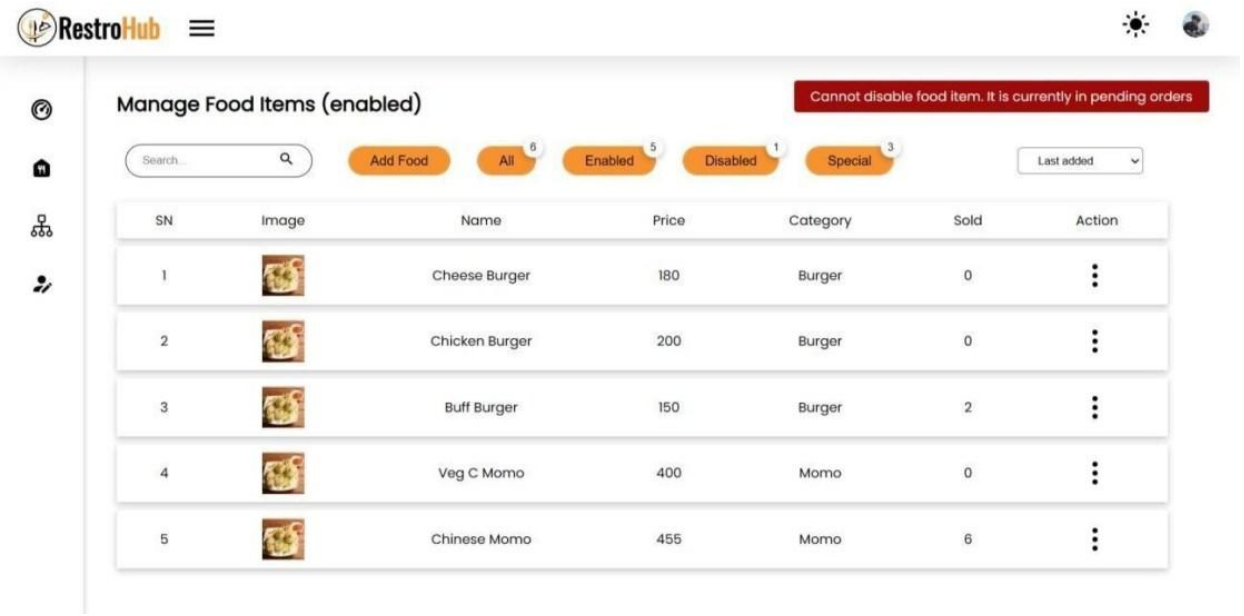


Figure 42: Food management page

When admin disable food items it gives an message as shown in above figure.

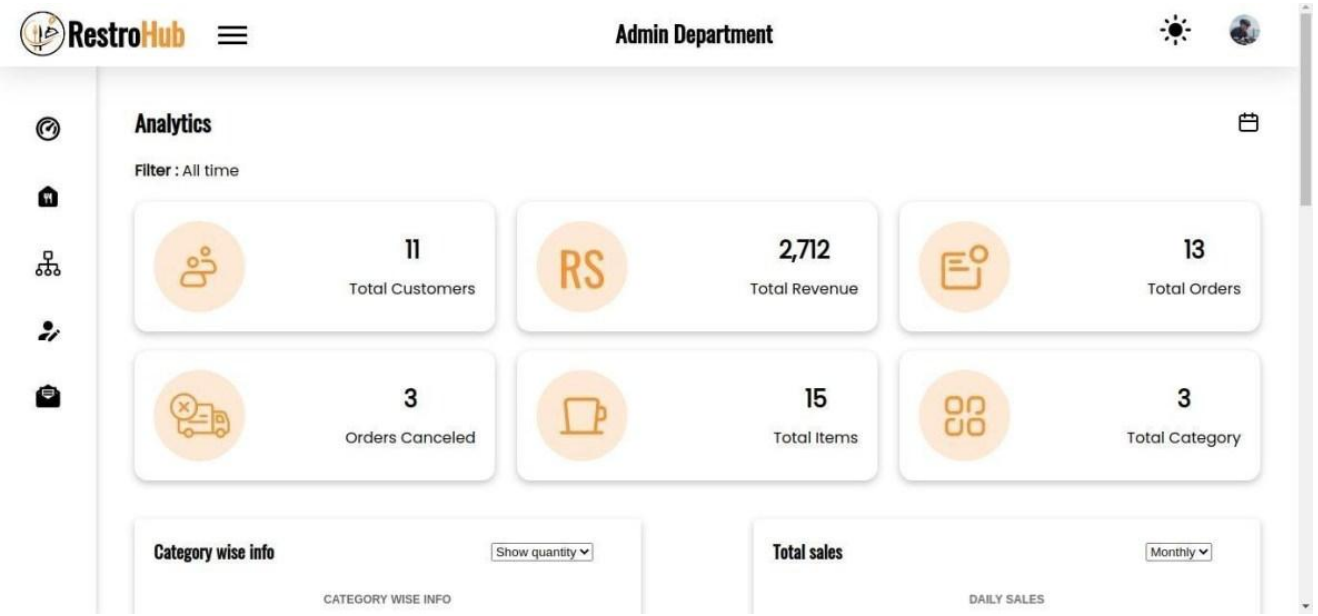


Figure 43: Analytics page (admin)

Admin can analyze all the details about customers, orders, revenue and sales of food from this page.

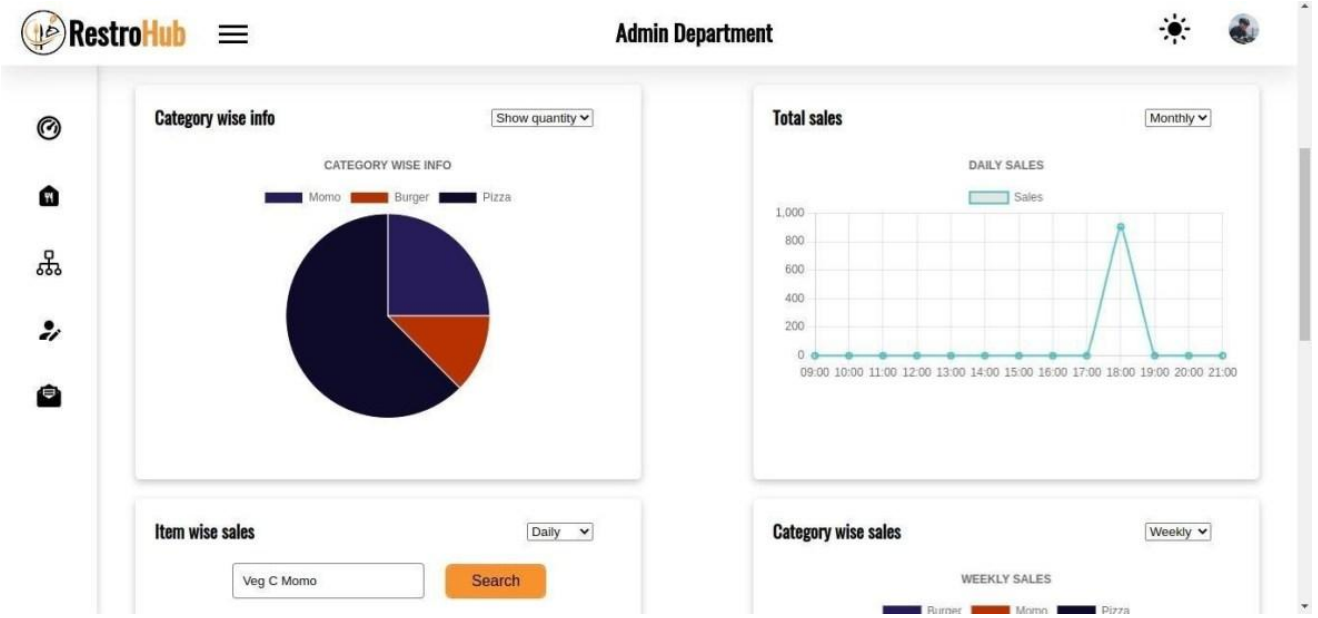


Figure 44: Analytics page (admin)

Admin can analyze details of total sales of food from this page

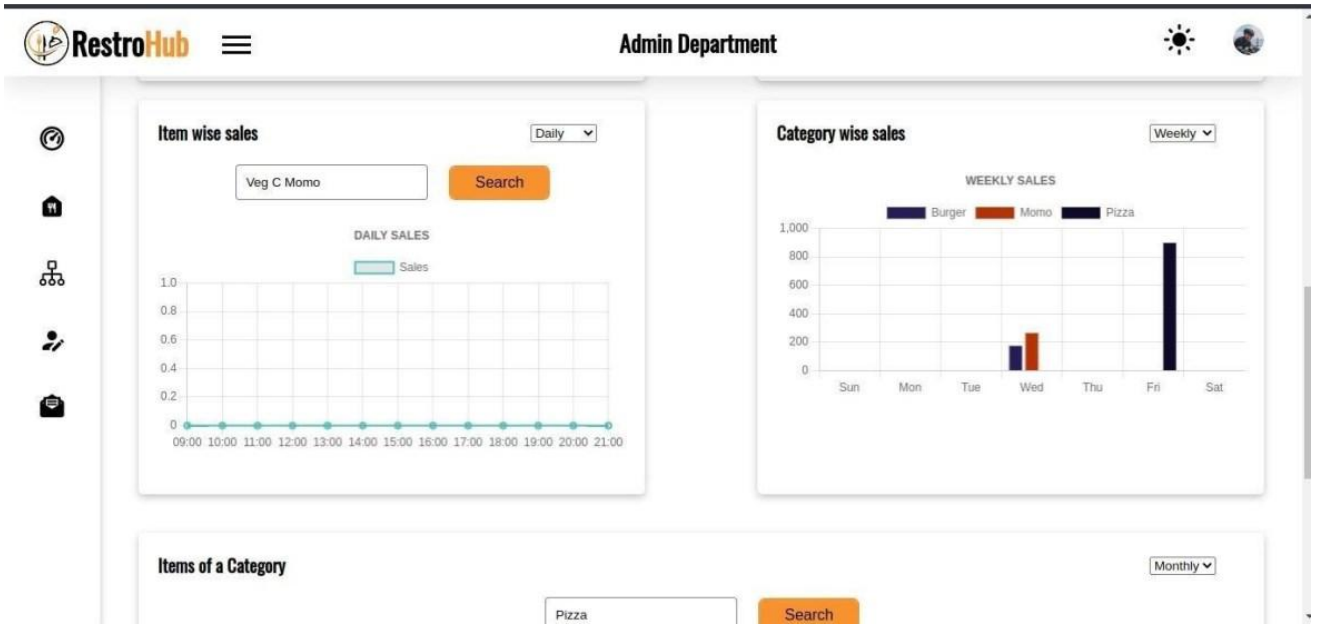


Figure 45: Analytics page (admin)

Admin can analyze item wise and category wise sales from this page

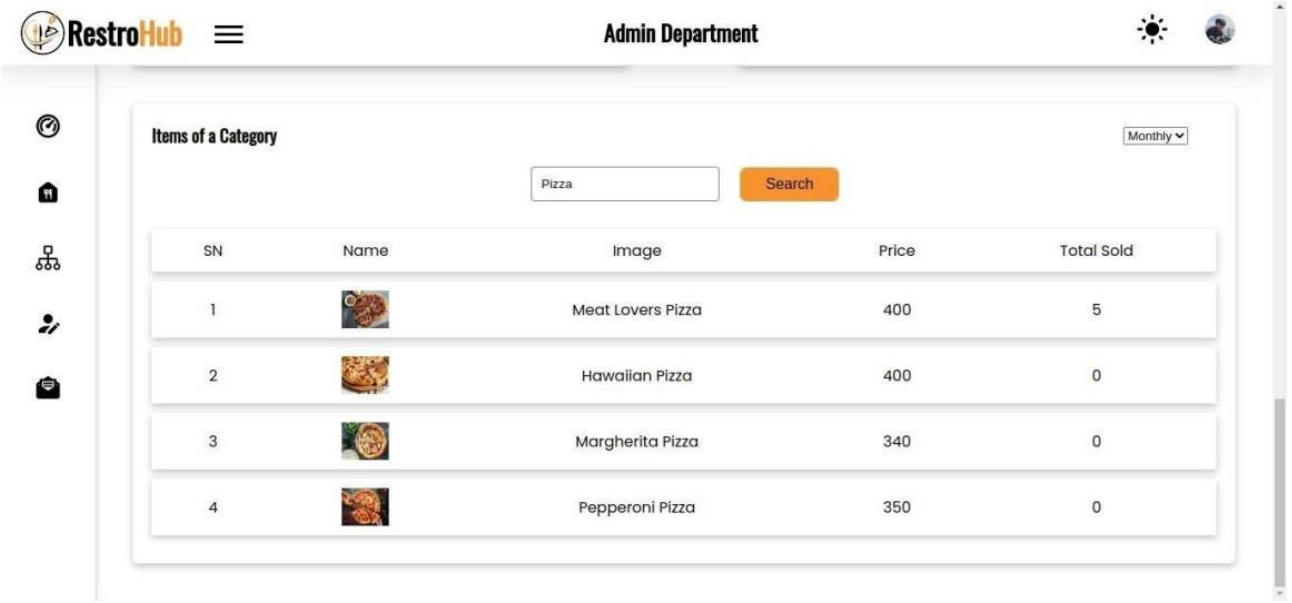


Figure 46: Analytics page (admin).

Admin can search category and see total sales of each category.

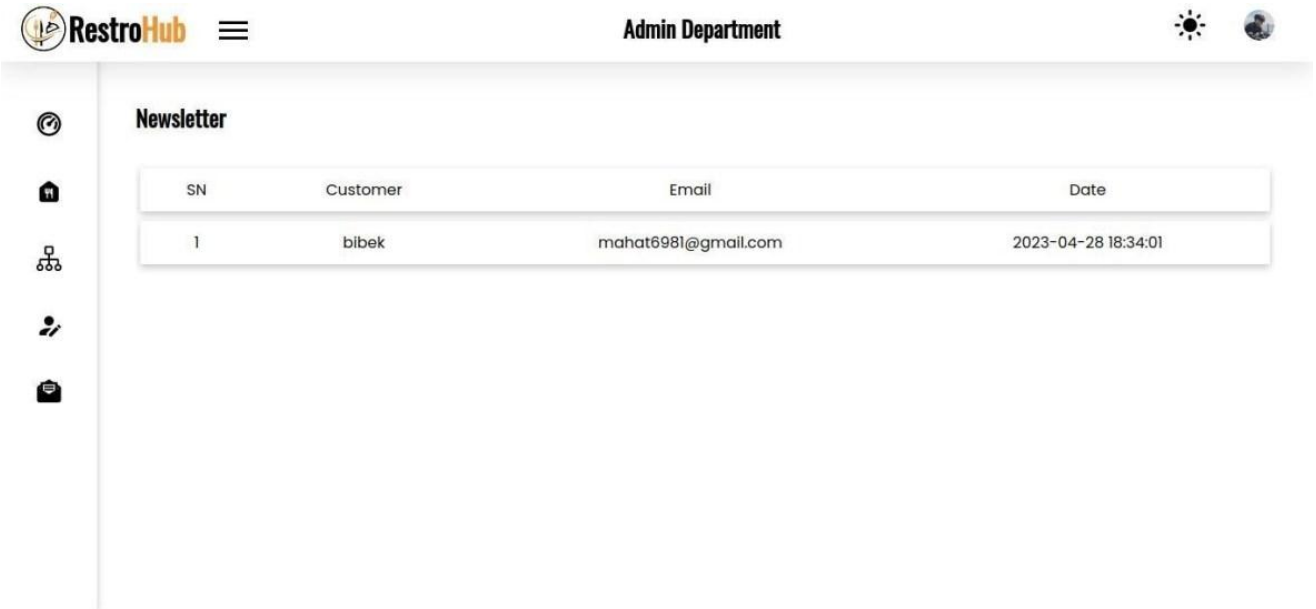


Figure 47: customer's stored email address page for newsletter (admin).

Admin can view customer's stored email address page for newsletter.

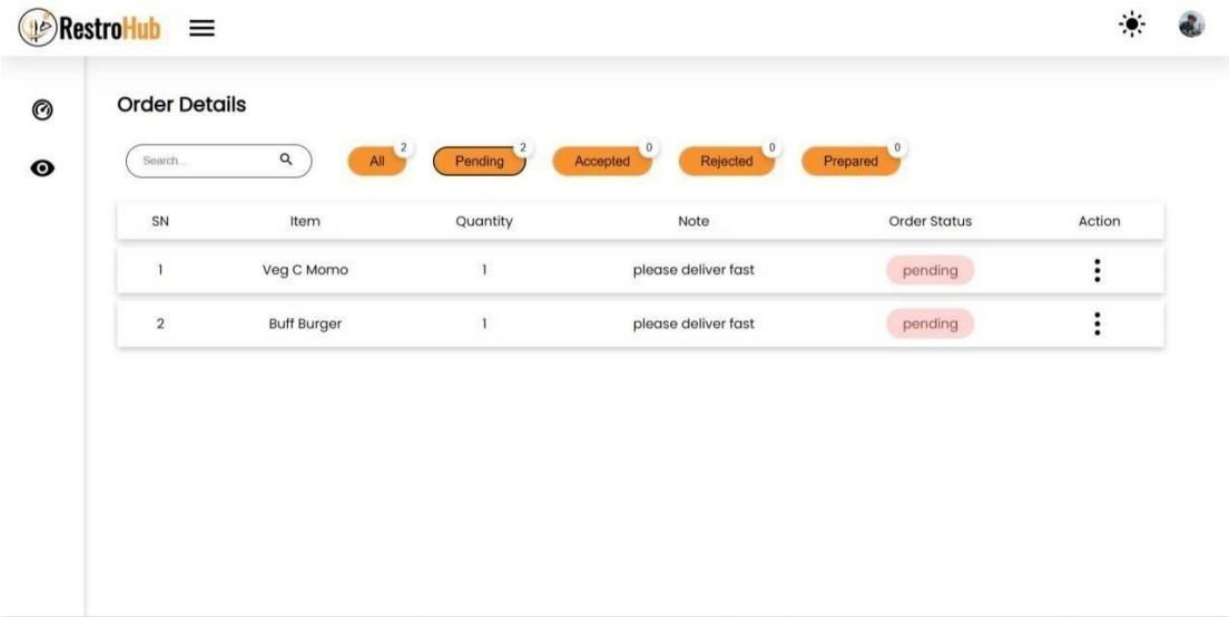


Figure 48: Order detail page (kitchen)

After successfully logging in, kitchen users are redirected to this page. From this page kitchen user can accept or reject orders.

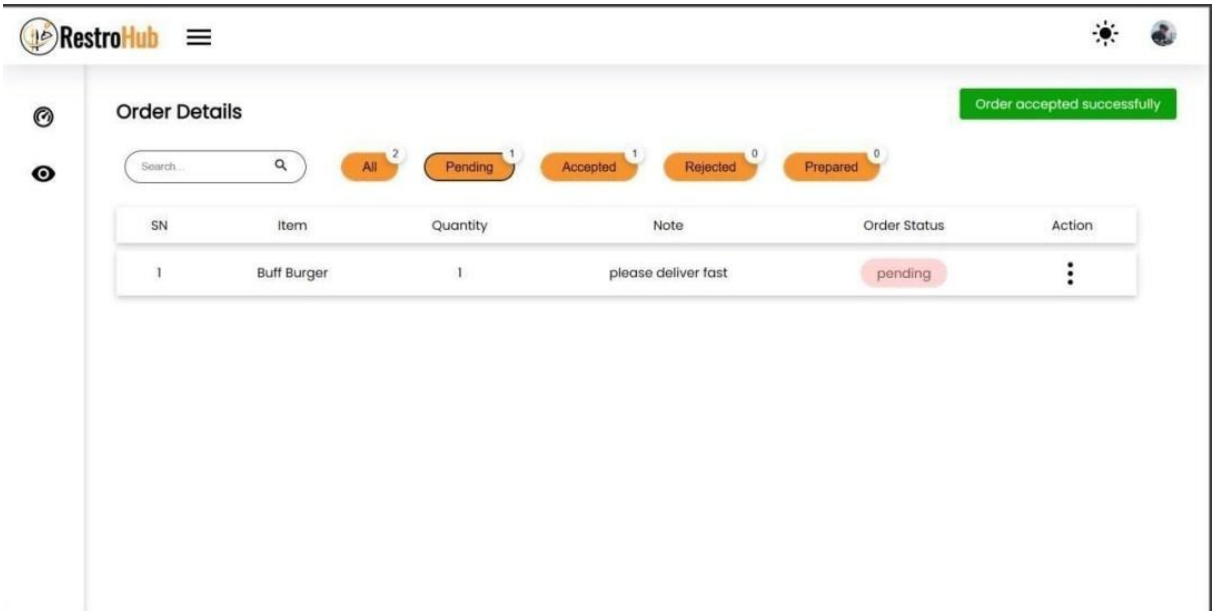


Figure 49: Order detail page (for kitchen)

When kitchen users accept the order, the order status is changed to pending.

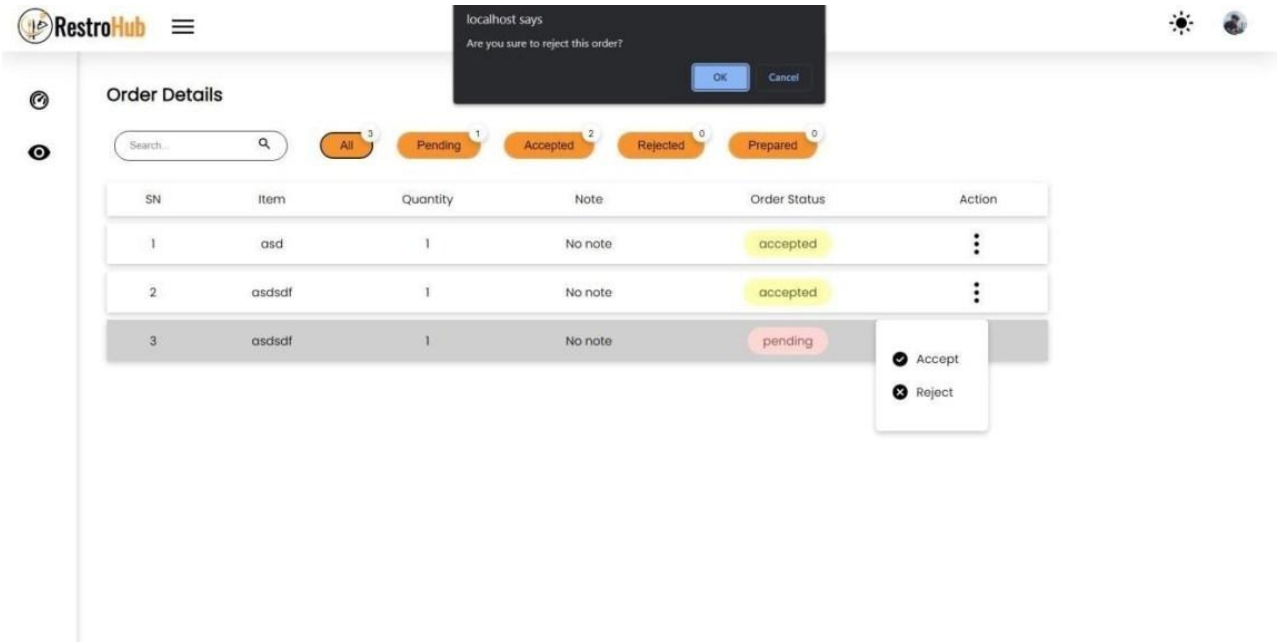


Figure 50: Order detail page (for kitchen)

When kitchen user clicks on “reject” button, it gives an alert an ask “Are you sure to reject this order”.

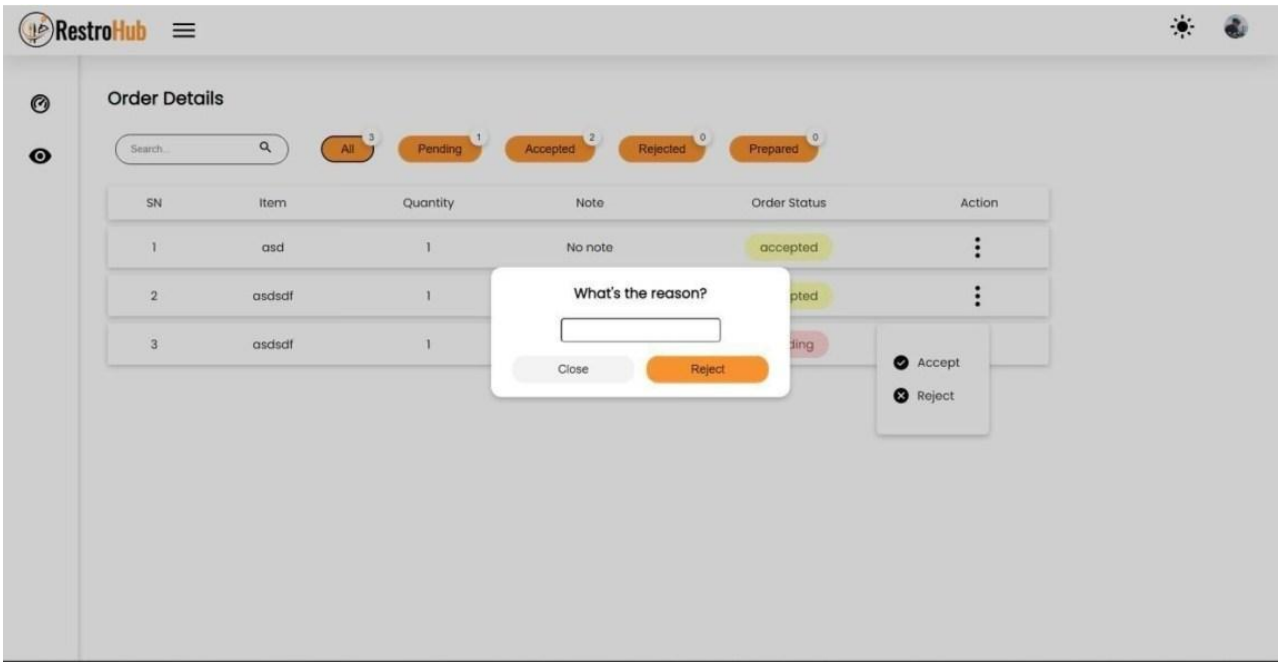


Figure 51: Order detail page (for kitchen)

When kitchen users reject the order, they must give the reason.

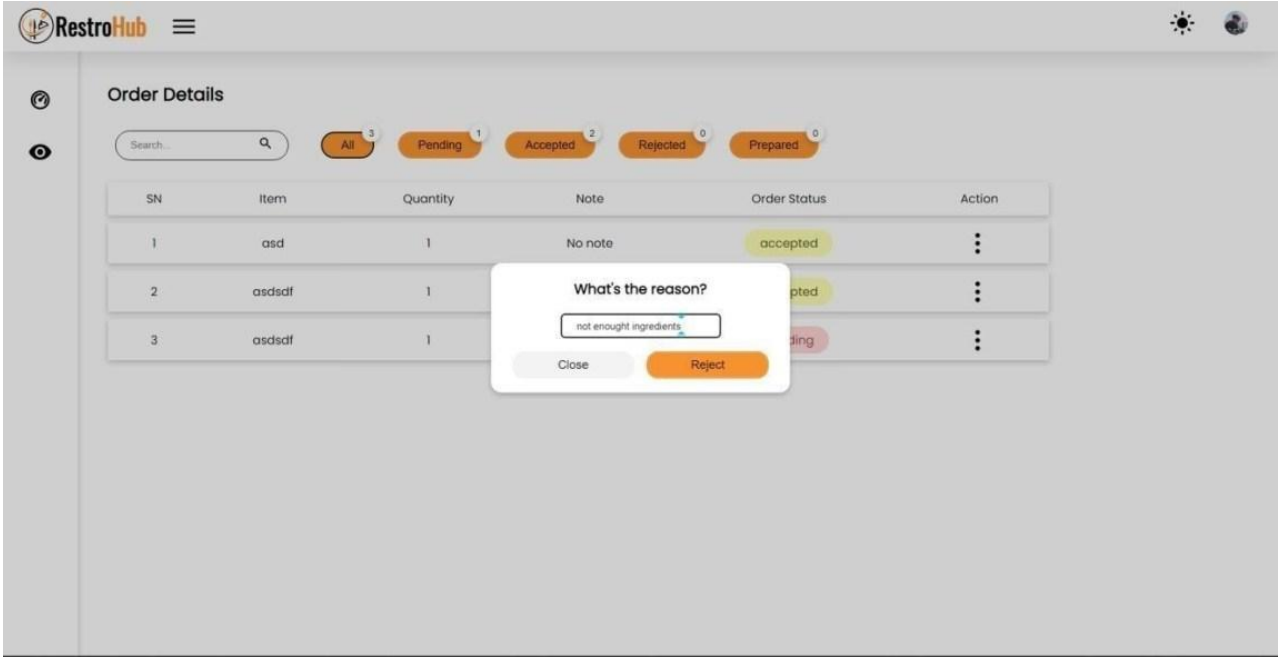


Figure 52: Order detail page (for kitchen)

Kitchen users have to give reason and click in the reject button.

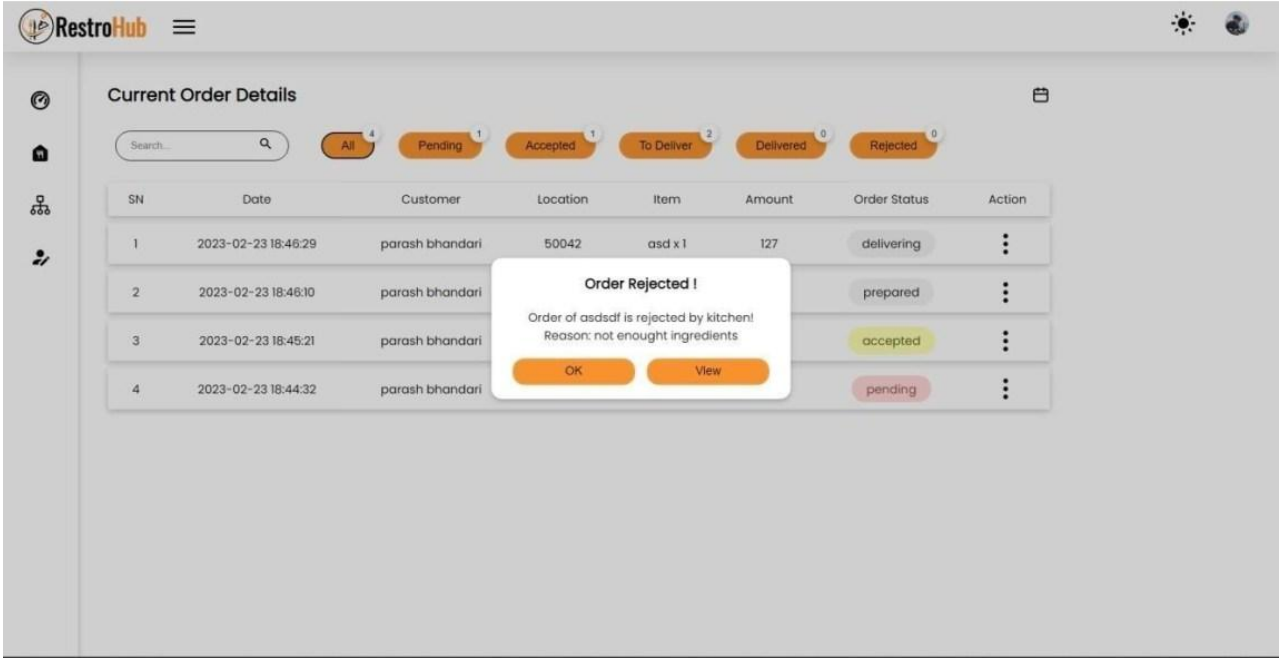


Figure 53: Order detail page (admin)

When kitchen users reject the order, a notification is sent to admin with the reason specified by kitchen user.

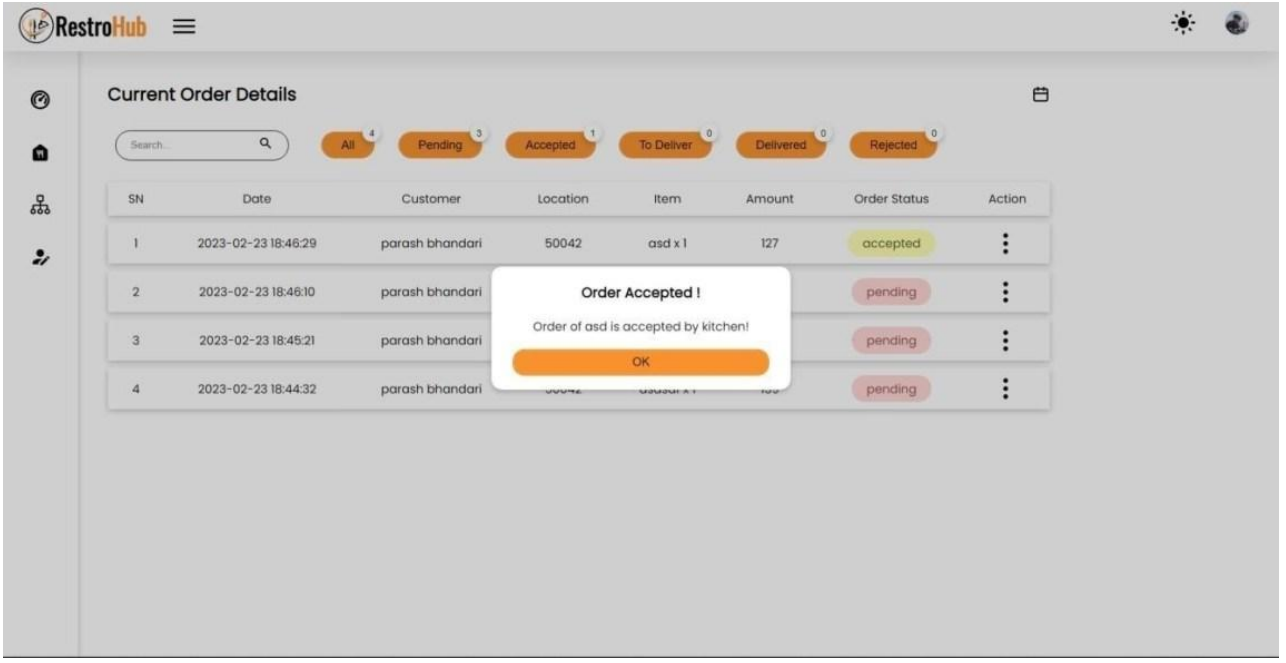


Figure 54: Order detail page (admin)

When kitchen users accept the order, a notification is sent to admin.

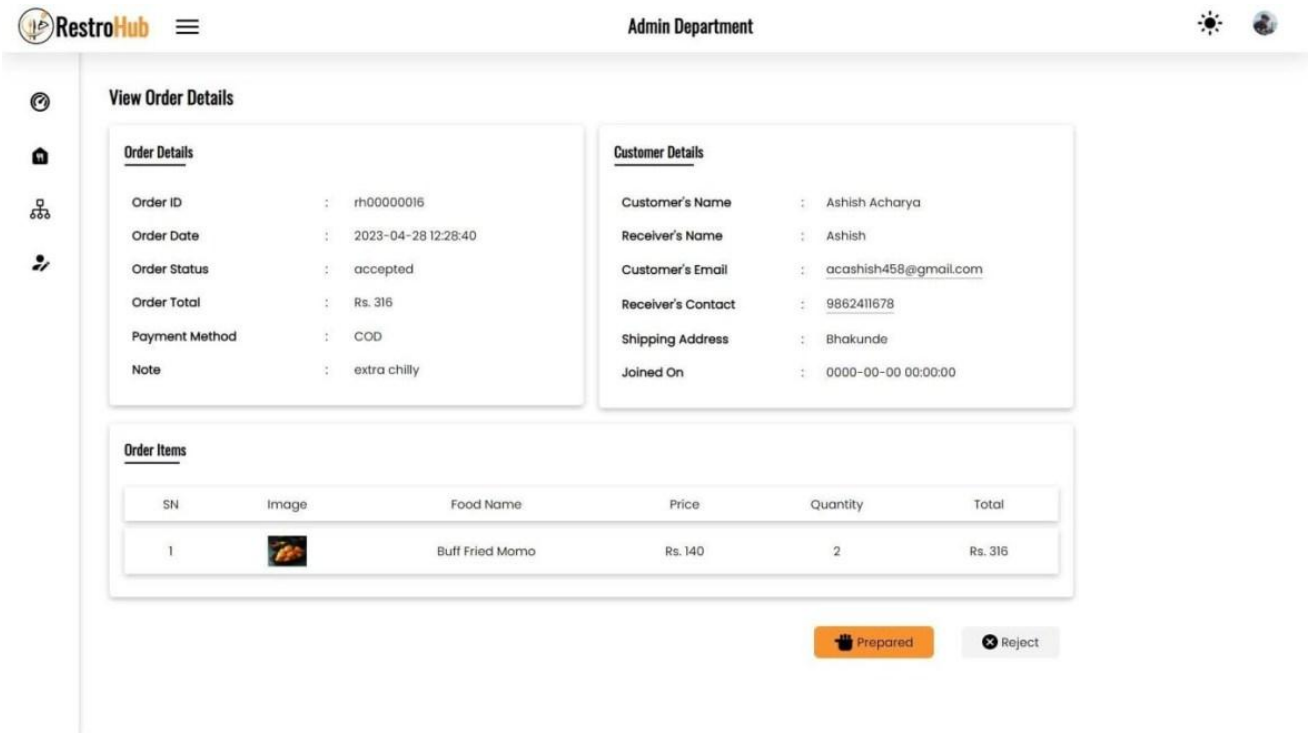


Figure 55: order details of customer (admin)

Admin can view details of order as well as all details of that user ordered food from this page.

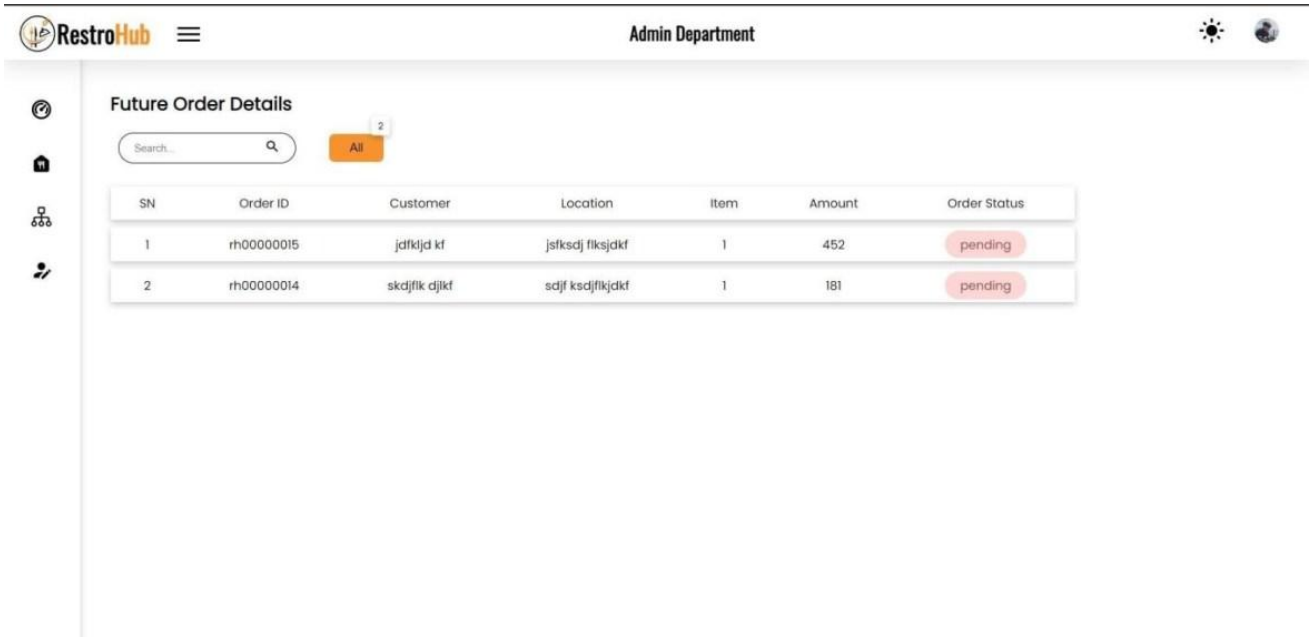


Figure 56: order detail page (admin)

Admin can view future order (order for later) from this page.

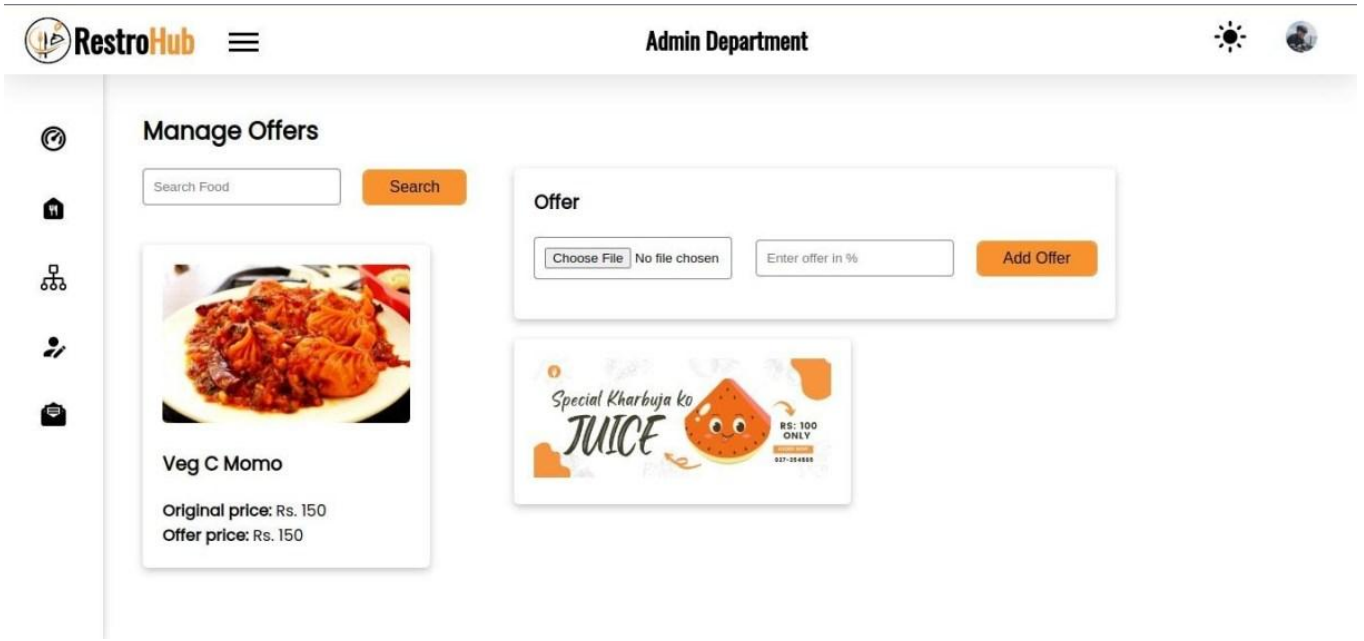


Figure 57: offer page (admin)

Admin can manage offers for customers from this page.

Figure 58: Login page (for delivery)

Delivery staff can login from this page.

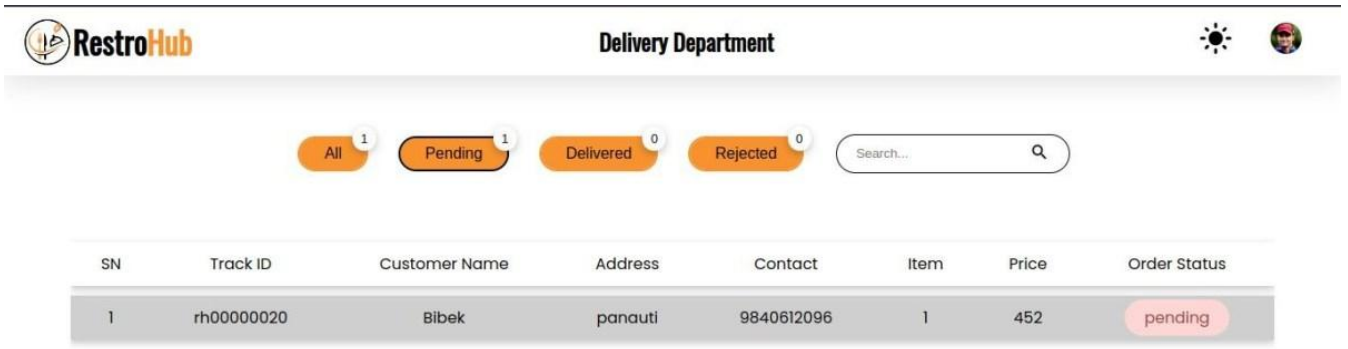


Figure 59: Index page (for delivery). Delivery staff has access to the list of prepared foods from this page.

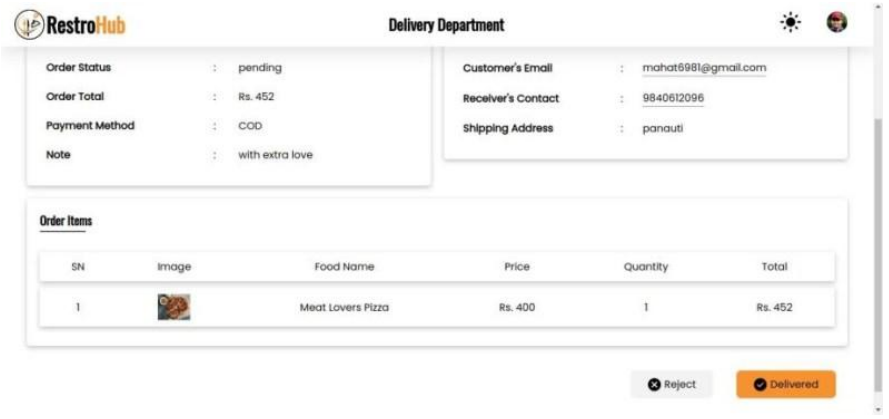


Figure 60: Confirmation page

Delivery staff can confirm the delivery of the order from this page.

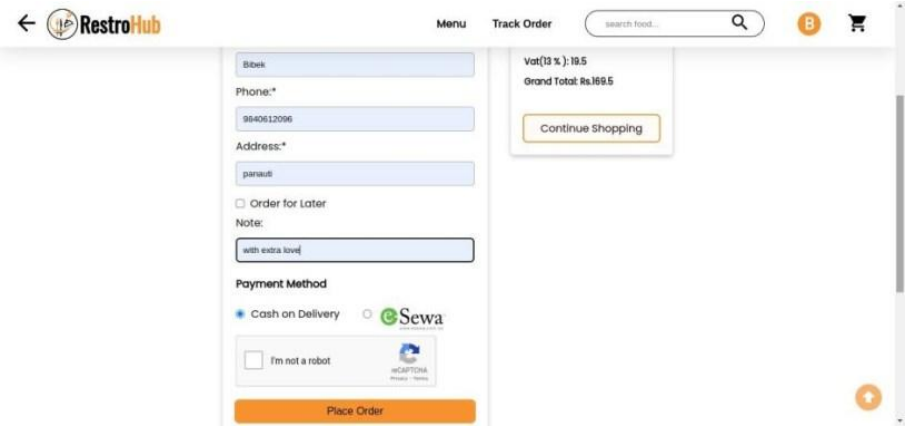


Figure 61: Captcha verification

To order food user have to complete capthcha verification.

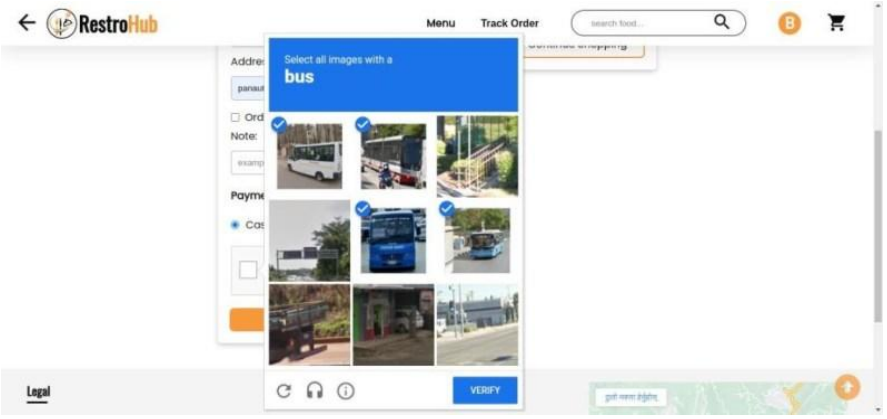


Figure 62: Captcha verification

8. CONCLUSION

- Customers are allowed to order food online.
- The menu of a restaurant is digitalized.
- Admin is allowed to view sales record easily.

9. FUTURE WORKS

- Authentication by Facebook is to be added.
- Total profit is to be calculated.

10. GANTT CHART

Week	1	2	3	4	5	6	7	8	9
Requirement									
Analysis									
Design									
Implementation									
Review									
Maintenance									
Documentation									

Figure 42: Gantt chart of online food ordering system

11. REFERENCE

- https://www.researchgate.net/publication/310020975_Online_Food_Ordering_System
- http://eprints.utar.edu.my/3448/1/fyp_IA_2019_WSJ_1506513.pdf
- https://ieeexplore.ieee.org/abstract/document/6089405)
- https://www.academia.edu/13345797/A_Proposed_System_for_Touchpad_Based_Food_Ordering_System_Using_Android_Application
- https://foodmandu.com/
- https://www.bhojdeals.com/
- https://www.zomato.com/